CDE4301 Innovation and Design Capstone

AY 26/27 Semester 1

Design and Evaluation of Modular Botanical Air-Treatment System

Interim Report

Group SSC-404

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Acknowledgements

We would like to thank our project supervisors, Dr Elliot Law, Dr Jovan Tan and Dr Veerasekaran S/O P Arumugam for their guidance and feedback throughout the project. We are grateful to Mr Richard for his technical advice and laboratory support. We also thank Dr Md Raisul Islam and Dr Guo Shuai for their guidance in using Ansys for our simulations.

Contents

Acknowledgements **2**

Abstract **5**

1. Introduction **6**

1.1. Significance of Indoor Air Quality 6

1.2. Air Pollutants of Interest 6

1.3 Coexistence and Variation of Indoor Pollutants 7

1.4 Target Indoor Spaces and Selection Rationale 9

2. Review of existing approaches and project gaps **10**

2.1 Current IAQ Management Approaches and Limitations 10

2.2 Active Botanical Filtration and Supporting Research 11

2.3 Existing Botanical Filtration Systems 14

2.4 Challenges in Existing Botanical Systems 16

3. Project Definition **17**

3.1. Problem Statement and Objective 17

3.2. Project Scope 17

3.3. Key Design Considerations 18

3.4. Measurable Requirements 19

4. Concept Development **21**

4.1. Reference System and Plant 21

4.2. Modular Configuration Selection 23

4.3. Overall System Architecture..... 25

4.4. Concept Features Rationale..... 26

5. Detailed engineering design and development..... 27

5.1 Plant-module construction..... 27

5.2 Irrigation design..... 27

5.3 Airflow and treatment-performance considerations..... 29

5.4 Baffle development and CFD findings..... 30

5.5 Fan selection and control.....32

5.6 Lighting design and calibration..... 32

5.7 Base-module design..... 33

5.8 Sensor module and integration.....33

6. Validation plan..... 35

6.1 Validation objectives and performance metrics..... 35

6.2 Experiment 1: Fan flow-rate optimisation..... 35

6.3 Experiment 2: Room-volume analysis with a continuous source.....36

6.4 Experiment 3: Module-count and treatment-rate relationship.....37

6.5 Development of the sizing guide.....38

7. Progress and next steps..... 40

7.1 Prototyping progress.....40

7.2 Mapping progress to the objectives..... 40

7.3 Outstanding technical questions.....41

7.4 Remaining work and project sequence.....42

References..... 44

Appendix..... 50

Appendix : Footprint Benchmark Using Naava One..... 50

Appendix : SPRE and CADR Definition..... 52

Appendix : Flora Aer Reference-System Measurements..... 53

Appendix Development of the module interlocking mechanism..... 54

Appendix Baffle Simulation Comparison..... 57

Appendix Lighting metrics and supporting calculations..... 62

Appendix Experiment 1 — Fan flow-rate optimisation..... 63

Appendix 7.1 Experimental procedure..... 63

Appendix 7.2 Airflow calculation..... 63

Appendix 7.3 SPRE and single-pass CADR..... 64

Appendix 7.4 Operating-point selection..... 64

Appendix : Experiment 2 — Room-configuration analysis with a continuous source..... 65

Appendix 8.1 Experimental procedure..... 65

Appendix 8.2 Baseline definition..... 65

Appendix 8.3 Continuous-source mass balance..... 65

Appendix 8.4 Equilibrium concentration..... 66

Appendix 8.5 Time-dependent concentration..... 66

Appendix 8.6 Effective treatment rate..... 67

Appendix 8.7 Fitting and interpretation..... 67

Appendix 8.8 Supplementary placement and mobility assessment..... 68

Appendix : Experiment 3 — Module-count and treatment-rate relationship..... 69

Appendix 9.1 Experimental procedure..... 69

Appendix 9.2 Treatment-rate calculation..... 69

Appendix 9.3 Scaling efficiency..... 70

Appendix : Sizing-guide calculations and interpretation..... 72

Appendix 10.1 Room-volume calculation.....72

Appendix 10.2 Equivalent air changes per hour..... 72

Appendix 10.3 Required treatment capacity..... 72

Appendix 10.4 Module-count projections.....73

Appendix 10.5 Limits of the sizing guide.....73

Abstract

Small enclosed workplaces with limited natural ventilation can experience recurring exposure to particulate and gaseous pollutants, while restricted floor space complicates the installation of air-treatment systems. This project investigates a modular active botanical air-treatment system for workspaces of approximately 100–300 ft² (9.3–27.9 m²), intended to supplement ventilation and source control. Using Flora Aer (Botanical Filtration prototype) as a reference and retaining philodendron scandens, the proposed design integrates repeatable plant modules, supporting base modules and a separate air-quality sensor module. Fan-assisted airflow through the planted growing medium, distributed lighting and automated irrigation support treatment operation, while sensor feedback is intended to enable pollutant monitoring and threshold-based operation control.

Work completed includes concept selection, detailed module design, component shortlisting, preliminary mechanical prototyping and qualitative computational fluid dynamics comparisons. Three airflow configurations were investigated, with the second perforated-baffle design showing a more uniform simulated airflow distribution across the pot openings and being selected for prototype evaluation. These findings remain preliminary; pollutant-removal performance and compliance with design requirements have not yet been experimentally verified.

The next phase will assemble and verify the functional prototype before conducting experiments on fan operating settings, room configuration and module count. Measurements of pollutant-specific removal efficiency, clean air delivery rate and effective room treatment rate will inform design refinement and a sizing guide linking module configurations to treatment capacity within the tested operating range.

1. Introduction

1.1. Significance of Indoor Air Quality

Indoor Air Quality (IAQ) describes the condition of air within buildings in relation to the people who occupy them. IAQ is assessed by measuring airborne pollutant concentrations and environmental parameters such as temperature, relative humidity and comparing the measurements with the relevant recommended limits (NEA, n.d.). Good IAQ is crucial to human health and well-being, with implications for respiratory and cardiovascular health as well as cognitive performance (EPA, 2026; Allen et al., 2016). However, indoor air pollution remains a major yet overlooked concern, while outdoor sources such as urban traffic and industrial emissions receive greater attention. A 2025 World Health Organization (WHO) publication cited an estimated 3.2 million deaths annually associated with household air pollution from polluting cooking fuels (Cowell & Singh, 2025). More broadly, daily activities such as cooking, heating and cleaning can release pollutants that accumulate when ventilation is inadequate, often without visible signs (EPA, 2026).

In Singapore, the hot and humid climate encourages reliance on air-conditioning in indoor environments such as shopping malls, homes, schools and offices. People in Singapore have been estimated to spend approximately 90% of their time indoors (Qing, 2023). In these spaces, insufficient outdoor air-ventilation can allow pollutants from daily activities, furnishings and equipment to accumulate despite the room being cooled. Dyson's connected-purifier study found that Singapore's average indoor PM2.5 concentration in 2022 was 36% higher than outdoor levels, ranking fourth among the countries studied for this disparity (Dyson, 2024). These findings reinforce the importance of maintaining good IAQ. Local concerns were further highlighted in a 2024 report describing a family whose children repeatedly fell ill in a home with

elevated formaldehyde levels associated with wooden furnishings (Seah & Tang, 2024). This underscores the relevance of the National Environment Agency's guidance on maintaining adequate ventilation and adapting IAQ management measures to each space's occupancy, activities and environmental conditions (NEA, 2026).

1.2. Air Pollutants of Interest

Indoor environments contain a range of airborne pollutants originating from occupants, activities, materials and outdoor air. This project focuses on four air quality parameters: carbon dioxide (CO_2), fine particulate matter (PM2.5), total volatile organic compounds (TVOCs) and formaldehyde. Table 1 summarises their common sources, acceptable limits and corresponding average periods to guide the interpretation of measured concentrations.

Table 1: Common sources and acceptable limits for the selected air quality parameters (SS 554:2016+A1:2021, Annex C.1, pp. 34–35; US EPA, "Indoor Particulate Matter" (n.d.).)

Parameters	Common sources	Acceptable Limits	Averaging periods
CO_2	Occupants' exhaled breath and combustion processes	700 ppm above outdoor concentration	8 hours
PM2.5	Cooking, tobacco smoke, candle burning and fine	37.5 µg/m³	24 hours

	particles entering from outdoors.		
TVOCs	Paints, coatings, adhesives, cleaning agents, air fresheners and personal-care products.	1,000 ppb, expressed as isobutylene equivalent	8 hours
Formaldehyde	Pressed-wood furnishings and certain insulation materials, cosmetics ,disinfectants and adhesives	100 μg/m³ (approximately 0.08 ppm)	30 mins

These four parameters were selected to represent the different aspects of air quality in the intended workspaces. CO_2 is used as a proxy for ventilation adequacy when interpreted in relation to room occupancy and the outdoor CO_2 concentration (NEA, 2022). PM2.5 represents the fine-particle pollution and TVOCs provide an aggregate indication of gaseous emissions from indoor products and activities. Although formaldehyde is a VOC, it is monitored separately because of its relevance from furnishings and building materials as well as because TVOC reading alone does not establish its individual concentration.

Monitoring these parameters allows the project to examine how the indoor conditions vary with occupancy, activities and material emissions. Their coexistence and variation over time are discussed in the following section.

1.3 Coexistence and Variation of Indoor Pollutants

Indoor air pollutants can coexist within the same space because several sources may be present simultaneously. Nail salons would be one such example. Occupants release CO_2 through respiration, nail-care products emit VOCs, and nail filing generates airborne dust. Certain nail-care products may also emit formaldehyde, depending on their composition (OSHA, n.d.). These sources contribute to a mixture of gaseous and particulate pollutants within the same indoor environment.

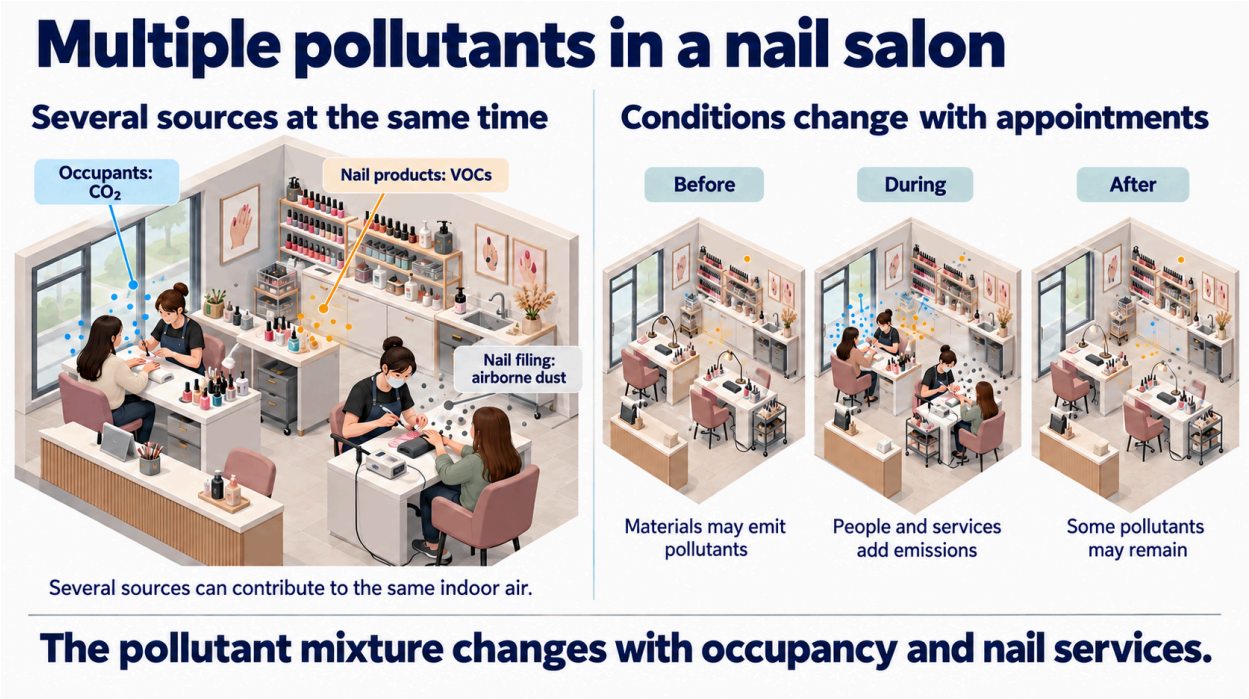

Figure 1: Pollutant sources and changes in the pollutant mixture over time in a nail salon

Note: Illustration generated using ChatGPT (OpenAI, 2026)

As illustrated in Figure 1, this mixture can change before, during and after appointments. Before an appointment, furnishings and other indoor materials may already contribute background emissions. During appointments, the presence of customers and staff, application of nail products and filing activities introduce additional emissions. After appointments, some pollutants may remain while others decrease as they are removed through ventilation or other processes. The resulting concentrations therefore depend on the activities performed, occupancy and ventilation conditions and need not rise or fall together.

This variation motivates monitoring multiple air quality parameters over time rather than relying on a single measurement. For this project, monitoring CO_2, PM2.5, TVOCs and formaldehyde is intended to capture different aspects of changing indoor conditions and inform the system's response.

1.4 Target Indoor Spaces and Selection Rationale

The nail salon scenario discussed in Section 1.3 motivated the investigation of other indoor spaces with similar conditions. The project therefore adopted the following three screening criteria for potential applications:

- limited natural ventilation
- recurring sources of multiple pollutants
- prolonged occupancy of approximately 5 - 6 hours per day.

These criteria focus attention on spaces where occupants may encounter emissions repeatedly throughout their working day.

To explore potential applications, preliminary desk research assisted by Google's AI search was conducted to identify candidate room types and estimate their numbers across different

floor-area ranges in Singapore. Figure 2 summarises these exploratory estimates. The investigation highlighted print and photocopy rooms and small discussion rooms within the 101–200 ft² category, as well as art and design studios and nail salons within the 201–300 ft² category. The estimates provided an initial basis for exploring possible applications.

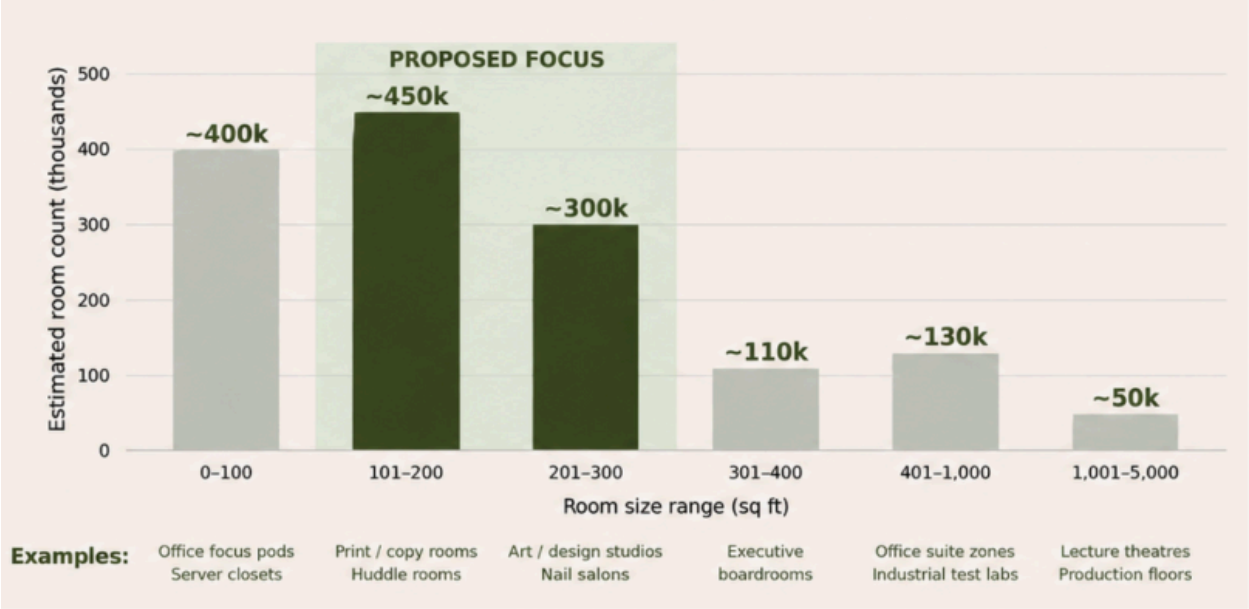

Figure 2: Preliminary estimates of candidate room numbers by floor-area range in Singapore

Following this investigation, a working scope of approximately 100–300 ft² (9.3–27.9 m²) was selected because the candidate room types identified within this range appeared to align most closely with the three screening criteria. These spaces were therefore prioritised for further investigation, with individual rooms to be assessed for their ventilation, recurring pollutant sources and occupancy duration.

2. Review of existing approaches and project gaps

2.1 Current IAQ Management Approaches and Limitations

The existing approaches to managing indoor air quality include source control, outdoor-air ventilation and air cleaning. Their effectiveness depends on the pollutants being addressed and the conditions within the space. Table 2 summarises the approaches relevant to the four air quality parameters considered in this project.

Table 2: Current IAQ management approaches and its limitations (EPA, 2018; NEA, 2026)

Parameter	Current approaches	Main limitations
CO_2	Outdoor-air ventilation through heating, ventilation and air-conditioning (HVAC) systems	In hot and humid conditions, additional outdoor air increases the cooling and dehumidification load.
PM2.5	Air purifiers containing high efficiency particulate air (HEPA) filters	Performance depends on airflow, equipment sizing, placement and filter maintenance.
TVOCs	Source control, HVAC systems and gas-phase filtration using	Removal varies between compounds and filter media. Adsorbents have finite capacity and require replacement.

	activated carbon or other suitable adsorbents	
Formaldehyde	Lower-emission materials, outdoor-air ventilation and media specifically suited to formaldehyde removal	Existing materials may continue emitting formaldehyde. Removal performance depends on the treatment method, media capacity and operating conditions.

As illustrated in Figure 3, HVAC systems with outdoor-air ventilation help dilute CO_2 and TVOCs generated indoors. Part of the return air is exhausted, while the remainder mixes with incoming outdoor air before being filtered, cooled and supplied back to the room. When pollutant generation increases, greater outdoor-air exchange may be needed to maintain indoor concentrations. In Singapore's hot and humid climate, the additional outdoor air requires cooling and dehumidification, which can increase energy demand (Lawrence Berkeley National Laboratory, n.d.).

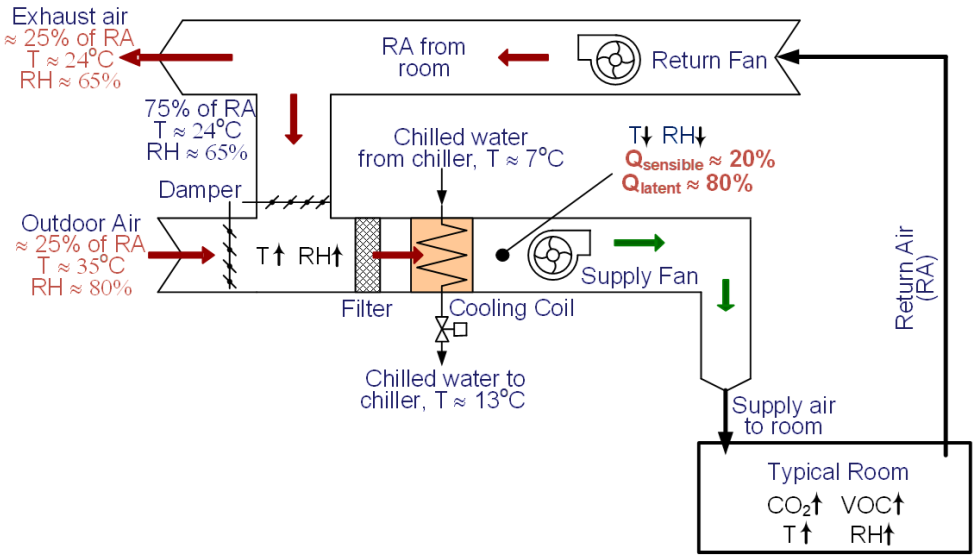

Figure 3: Typical airflow arrangement in a central HVAC system

Note: Diagram provided by Dr Md Raisul Islam

Air cleaning can supplement ventilation through HEPA filters for PM2.5 and suitable adsorbents, such as activated carbon, for selected gaseous pollutants. Their effectiveness depends on airflow, the pollutants present and filter maintenance (EPA, 2018). These approaches therefore require consideration of treatment performance, energy demand, maintenance and available space.

2.2 Active Botanical Filtration and Supporting Research

The approaches discussed in Section 2.1 highlight the need to balance pollutant removal with energy demand and maintenance requirements. These considerations motivate the investigation of botanical filtration as a supplementary treatment method. By combining plants, their growing medium and associated microorganisms, botanical systems offer the potential to address multiple pollutants within a single system (Matheson et al., 2023).

Figure 4 illustrates the complementary processes involved in phytoremediation, which uses plants and their associated microorganisms to remove or break down pollutants. Leaves can take up certain gaseous pollutants, while the growing medium can retain airborne particles and adsorb some gases. Microorganisms associated with plant roots can biodegrade certain organic pollutants. Plants also absorb CO_2 through photosynthesis. The contribution of each process therefore varies with the pollutant and growing conditions (Matheson et al., 2023).

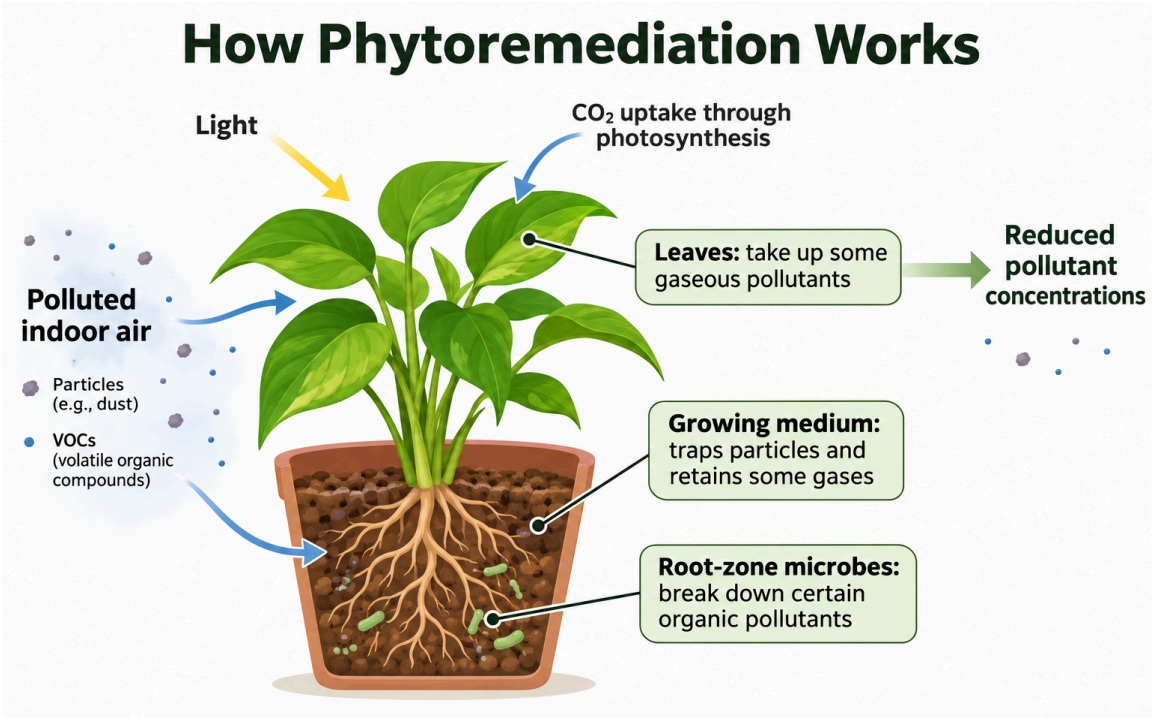

Figure 4: Phytoremediation Process

Note: Illustration generated using ChatGPT (OpenAI, 2026)

Active botanical systems incorporate fans to increase contact between polluted air and the treatment medium. Figure 5 shows a configuration in which a fan draws air into the base and directs it upwards through the central structure. Air then passes through the growing substrate within the surrounding planter modules before returning to the room (Smith et al., 2023). Unlike passive potted plants, this arrangement provides forced airflow through the planted medium, bringing pollutants into contact with the substrate and root-associated microorganisms.

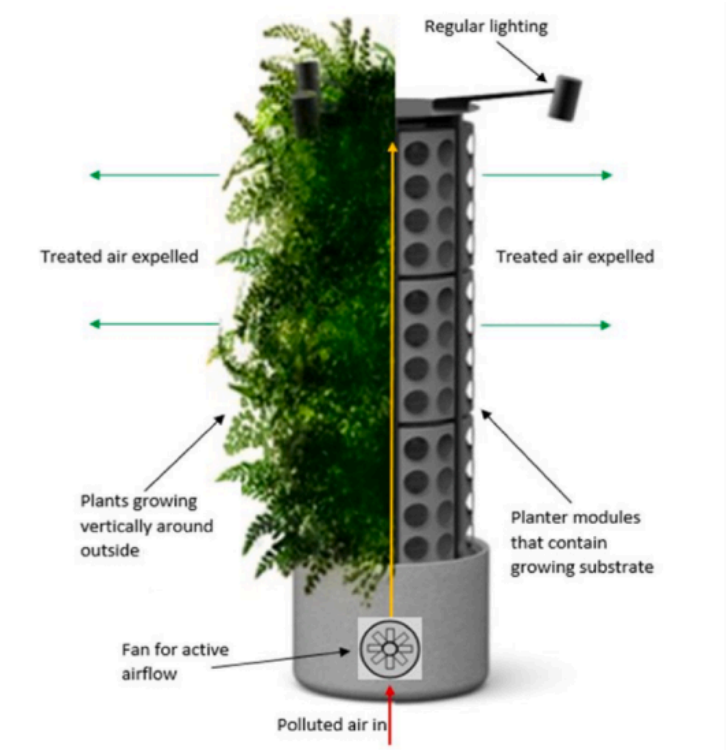

Figure 5: Active Botanical Filtration (Smith et al., 2023)

Experimental research supports the potential of this approach. Pongkua et al. (2024) investigated an active living wall planted with Sansevieria trifasciata in a 24 m³ chamber using cigarette smoke as the pollutant source (Figure 6). The authors reported more than 90% particulate matter removal within 12 hours, while removal of individual VOCs ranged from approximately 25–80%, including approximately 80% for formaldehyde. These findings demonstrate the potential to combine particulate and gaseous pollutant treatment, although the reported efficiencies reflect the study's experimental conditions and require further validation for the intended workspaces.

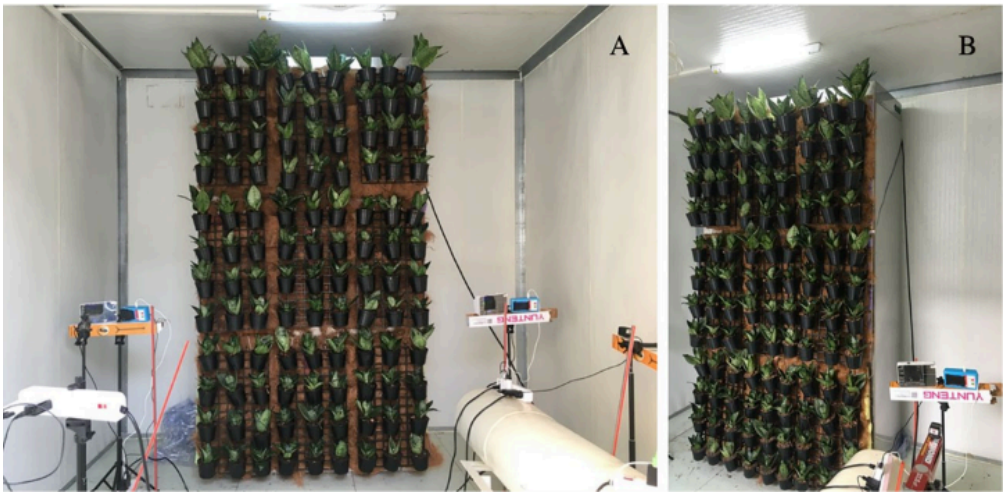

Figure 6: Active living wall in 24 m³ test chamber (Pongkua et al., 2024)

Beyond pollutant removal, botanical systems can provide a biophilic element by bringing living plants into indoor environments. Nieuwenhuis et al. (2014) found that enriching offices with plants improved workplace satisfaction and self-reported concentration. Combining these potential benefits with multiple pollutant-treatment mechanisms makes botanical filtration relevant to the selected workspaces. It is therefore investigated as an integrated supplementary approach.

2.3 Existing Botanical Filtration Systems

The botanical filtration principles discussed in Section 2.2 have been incorporated into commercial systems. Naava and Nedlaw illustrate two approaches to integrating plants and engineered airflow into indoor environments, with different configurations and installation requirements.

Naava, a Finnish company, offers freestanding vertical green walls that combine botanical air treatment with indoor greenery. These units can also serve as room dividers, allowing them to be incorporated into spaces such as offices, schools and public areas (Figure 7). Fans circulate

air through the planted growing medium, while irrigation and lighting support plant growth. This furniture-like configuration integrates botanical filtration into the layout of an occupied room (Naava, n.d.).

Figure 7: Naava green wall used as space divider (Sponda, 2022)

Nedlaw's custom-designed living-wall biofilters illustrate a larger-scale approach, with installations in atriums and institutional buildings (Figure 8). Air is drawn through the planted wall using onboard fans or a connection to the building's HVAC system. Each installation is designed around requirements such as available space, airflow and lighting, enabling the biofilter to form part of the building's air-treatment infrastructure (Nedlaw Living Walls, n.d.).

Figure 8: Multi-storey living wall integrated into an indoor atrium (Terra Landscaping, n.d.).

These examples demonstrate how botanical filtration can be incorporated into both freestanding units and architectural installations. Their differing configurations provide a basis for considering the space requirements, installation arrangements and maintenance needs relevant to the small enclosed workspaces targeted in this project.

2.4 Challenges in Existing Botanical Systems

The systems discussed in Section 2.3 demonstrate the practical application of botanical filtration. However, their suitability for small enclosed workspaces requires further consideration. Four challenges are particularly relevant to this project: deployment flexibility, performance observability, response to changing pollutant levels and room-scale performance.

Limited deployment flexibility: Large, custom-designed living walls require installation arrangements suited to the building, including sufficient space, lighting and airflow provision (Nedlaw Living Walls, n.d.). Freestanding systems offer flexibility in placement, but their

dimensions and treatment capacity must still suit the room. For the selected 100 - 300 ft² workspaces, the challenge is to accommodate different layouts and treatment requirements while preserving usable floor area.

Limited performance observability: Users need accessible information to understand how indoor pollutant levels change during operation. However, visual appeal can receive greater attention than air-cleaning performance. Shen et al. (2024) noted that interior green-wall design often prioritises aesthetics and cost, with insufficient consideration of plant species' VOC-removal capabilities. This highlights the need to support botanical systems' visual appeal with clearly presented measurements that help users assess their contribution to indoor air quality.

Unclear response to changing pollutant levels: Indoor emissions vary with activities and occupancy, as illustrated by the nail salon scenario in Section 1.3. A relevant question is how botanical systems adjust their operation when these conditions change. Although existing products mention incorporating automated functions, the relationship between measurements of the selected pollutants and adjustments to treatment airflow requires examination. The challenge is to determine an appropriate operating response to changing treatment demand.

Uncertain room-scale performance: Experimental removal efficiencies depend on factors such as pollutant concentration, airflow and test duration. Matheson et al. (2023) highlight the need for further assessment in actual indoor environments containing mixed pollutant sources. Performance in the selected workspaces therefore needs to be established under representative conditions, accounting for room volume, ventilation and recurring emissions. Such evaluation is necessary to determine whether the system provides a measurable contribution to indoor air treatment.

3. Project Definition

3.1. Problem Statement and Objective

Small enclosed workplaces with limited natural ventilation can experience recurring exposure to multiple pollutants released by activities, materials and occupants, raising concerns about health, comfort and work performance. Many individual treatment technologies are pollutant-specific, such as HEPA filtration for particulate matter and adsorbents for specific gases. Active botanical filtration was therefore selected for investigation because it offers the potential to address particles and certain gaseous pollutants within one integrated plant-substrate system.

However, its suitability for the intended spaces remains uncertain due to:

- Limited deployment flexibility: Fixed dimensions or positions can restrict deployment across room layouts and relocation nearer to pollutant sources.
- Limited visibility of IAQ performance: Insufficient accessible information makes changing pollutant levels and treatment effects difficult to assess.
- Unclear responsiveness: The relationship between sensor readings and system operation is not sufficiently established for the intended application.
- Uncertain treatment capacity: Performance under the intended room sizes, ventilation conditions and pollutant sources remains insufficiently verified.

The project therefore aims to help operators of enclosed workplaces with limited natural ventilation manage recurring indoor pollutants through adaptable botanical air treatment that displays changes in pollutant levels, responds to changing conditions and provides experimentally verified treatment capacity within a defined operating range.

3.2. Project Scope

The project covers the design, development and evaluation of an adaptable botanical air-treatment system for small enclosed workplaces with limited natural ventilation. The intended application range is 100–300 ft² (approximately 9.29–27.87 m²), with suitability to be assessed through testing under defined room and operating conditions.

Development includes the physical structure, airflow delivery and conditions needed to sustain plant operation, including lighting and irrigation. The work also covers monitoring for automation and displaying indoor air conditions. Building on an existing botanical treatment approach, the team's contribution focuses on engineering development, integration and experimental evaluation. Testing will investigate pollutant removal under different operating conditions and room sizes to establish treatment capacity and inform sizing guidance. The intended deliverables are a functional prototype, measured performance data and recommendations for use within the tested range.

Treatment evaluation will address PM2.5 and selected gaseous pollutants (TVOCs), including formaldehyde. CO_2, temperature and relative humidity will provide additional operating context. The system is intended to supplement ventilation and source control. Microbial strain development, comprehensive plant-species screening and demonstration of health outcomes are outside the project scope.

Future work will be divided by subsystem, with each lead responsible for development, testing and subsequent improvements in their area. This supports parallel progress through the prototype development and design refinement phases shown in the Gantt chart (see Figure 25). System integration and whole-system testing will be undertaken collaboratively.

Table 3: Workload Distribution

Area of responsibility	Planned future work	Primary lead
Airflow	Implement and test the airflow system, investigate suitable fan operating flow rate and refine it based on findings.	Shawn Tan
Lighting	Install and calibrate lighting, assess its distribution for the plants and make adjustments during prototype development.	Liu Yi Bin
Sensors and automation	Integrate and test sensing and control functions, then support checks of automation and reliability.	Shadman Sudad

3.3. Key Design Considerations

The four identified gaps are translated into the following design considerations that guide concept selection and detailed development. These considerations concern the suitability of the proposed system for its intended application; they do not imply that all existing botanical systems lack the corresponding functions.

Table 4: Design Considerations

Gaps	Design Consideration	Design implication
Limited deployment flexibility	Modular and Mobile	• Permit changes in size and arrangement by module number • Support easy relocation.
Limited performance observability	Visibility of air quality	• Present relevant measurements, trends and operating status clearly to users. • Distinguish changes in room readings to show evidence of removal by the device.
Unclear Responsiveness to changing pollutant levels	Feedback capability	• Activate treatment when either designated pollutant indicator exceeds its threshold, and stop treatment when the defined deactivation conditions are met
Uncertain room-scale performance	Verified treatment capacity	• Measure pollutant-specific removal and treatment capacity under defined operating and room conditions to support sizing guidance .

3.4. Measurable Requirements

Table 5 consolidates the current design targets and their basis. Literature and commercial products provide benchmarks, but performance reported for another plant-substrate system is not evidence that the proposed system already achieves the same result. Numerical targets remain provisional until their operating and measurement conditions are specified and tested. Single Pass Removal Efficiency (SPRE) is used as target purification performance, refer to Appendix 2 for detailed definition of the term.

Table 5: Measurable Requirements

Requirement	Working target	Basis
Assembly footprint	≤ 5% of room floor area	Space-allocation target, informed by current existing living wall (NAAVA one) dimensions. The model takes up approximately 3.78% of a 100 ft² space (see Appendix 1). A higher target was set to accommodate plants.
Assembly height	2.4m–2.6m	Minimum height is determined by BCA (2025, p. 15). Maximum height is a planning assumption based on 100–300 ft² indoor spaces in Singapore, not a universal permissible assembly height.
Module dimensions	L + W + H ≤ 150 cm (Linear Dimension)	• Based on Technical Reference (TR) 105:2022 (Guidelines for last mile delivery of parcels), which covers domestic parcels with combined length, breadth and height

		not exceeding 150 cm. This informs the module's packed size for individual delivery (Enterprise Singapore, 2022, pp. 4–5). ● According to Scoot's checked-baggage linear-dimension limit (≤ 158 cm) as a benchmark (FlyScoot, n.d.).
Module Weight	≤ 25 kg	● Singapore Standards (SS) 569:2011 (Code of practice for manual handling) recommends a maximum of 25 kg for short-duration, low-frequency lifting under ideal conditions, including keeping the load close to the body, good two-handed grip and no twisting. This informs an initial upper reference (Enterprise Singapore, 2011, p. 35). ● Additional support is drawn from TR 105:2022 (Guidelines for last mile delivery of parcels), which covers parcels weighing no more than 25 kg or one-third of the handler's body weight, whichever is lighter (Enterprise Singapore, 2022, pp. 4–5).
PM2.5 removal	SPRE ≥ 55%	Irga et al. (2023) reported PM2.5 SPRE of 56.42 ± 21.02% for a moveable active green wall. This is

		used as a comparison benchmark, recognising the reported variability and different systems.
TVOCs removal	SPRE ≥ 55%	Torpy et al. (2017) reported approximately 57% SPRE for an active living wall tested with methyl ethyl ketone (MEK), a VOC. This provides a comparison benchmark.
Formaldehyde removal	SPRE ≥ 30%	Wu et al. (2026) reported formaldehyde SPRE of 24.75–34.5% for an active green-wall module under different airflow conditions. A 30% target is selected within this reported range as a comparison benchmark.
Outlet air speed (Plant-facing)	≤ 1 m/s	• Franco et al. (2012) limited active living-wall inlet velocities to 1.0 m/s, noting potential mechanical damage to vegetation at higher speeds. • Sabeh (2024) also recommends 0.35–1.0 m/s for air movement across plants
Water-refill interval	≥ 30 days	Mobilane's LivePanel TANK is an irrigation system for indoor living walls, using an integrated reservoir and pump to provide watering for an advertised 4–6 weeks per fill (Mobilane, n.d.). This supports a practical refill-interval benchmark.

Operating noise	≤ 50 dB(A)	Retained from SS 553:2026 (Code of Practice for Air-Conditioning and Mechanical Ventilation in buildings) ambient noise recommendation. Choosing office and retail shop standards as a benchmark (Enterprise Singapore, 2026).

4. Concept Development

4.1. Reference System and Plant

The existing Flora Aer prototype serves as the reference system for this project. It combines fan-assisted airflow through a planted growing medium with supplementary lighting and mist irrigation. Figure 9 shows its planting face and rear arrangement, providing a practical reference for accommodating the plants and integrating the services needed to sustain their operation.

Figure 9: Flora Aer

The prototype uses Philodendron scandens, with supporting research also reported under Philodendron hederaceum. Suárez-Cáceres & Patricia (2021) tested a planted felt wall in a sealed chamber under formaldehyde exposure, recording about 83% reduction in TVOC readings after 15 minutes and 96% after one hour, relative to the empty-chamber baseline. These measurements used a TVOC detector rather than a formaldehyde-specific instrument. For $PM_{2.5}$, Patil & Patil (2026) investigated a passive green wall planted with *Philodendron scandens* in an occupied office. Average PM2.5 concentrations decreased from approximately

34 to 28 μg/m³, with a reported reduction of 15.7%. This provides supporting evidence from passive planting (non-active) rather than direct validation of fan-assisted filtration.

Figure 10: Philodendron Scandens

Adopting the Flora Aer plant system provides a consistent basis for engineering development. Reference measurements inform pot accommodation and planting spacing (see Appendix 3), while plant requirements guide lighting and irrigation.

4.2. Modular Configuration Selection

For module configurations, an interactive configuration visualisation was used to visualise different ways of designing our modular system.

Interactive Element: livingwall-l3kwdyjj.manus.space

Figure 11: Vibe-Coded Website Visualisation

The proposed system uses repeatable flat, square plant modules that can be combined into horizontal and vertical arrays. Configuration selection was guided by four principal criteria: manufacturability, transportability, inter-module connectivity and plant spacing design.

Table 6 presents a qualitative comparison of flat, square modules against other configurations. The comparison concerns their suitability for sheet-based manufacturing and assembly approach, it is not an experimental ranking of purification performance.

Table 6: Module Configuration Considerations

Criterion	Flat, square modules	Hexagonal, Chevron, Wavy or Cylindrical
Manufacturability	• Planar parts suit the planned laser-cutting process. • Repeated rectangular components simplify fabrication and assembly.	• Hexagonal and chevron forms remain compatible with laser cutting but require additional angled-joint detailing. • Curved designs require forming processes as raw materials are usually flat.
Transportability	• Flat housing parts can be stacked before assembly. • Completed modules pack regularly for transport.	• Packing depends on the geometry and whether components can nest or separate. • Curved forms require additional consideration for compact transport.
Inter-module connectivity	• Repeated straight edges provide consistent alignment and fastening locations. • Horizontal and vertical connections simplify irrigation and electrical routing.	• Angled or curved interfaces require additional joint detailing. • Multiple connection directions for hexagonal form complicate standardising service connections.

Plant spacing design	• A regular grid simplifies maintaining consistent plant spacing within and across adjoining modules.	• Angled boundaries, recesses and curvature require more careful plant positioning to maintain spacing and canopy clearance across modules.

Flat, square modules were selected because they provide the most practical route to repeatable fabrication and assembly within the current project resources. The repeated geometry supports individual-module replacement. Permitted array sizes will also depend on design requirements (see Table 5), irrigation capacity, and testing of individual modules.

4.3. Overall System Architecture

The system is organised into three interacting subsystems: plant modules, base modules and a separate sensor module. Functions directly associated with each planting face are repeated within the plant modules, while support and shared services are concentrated in the base.

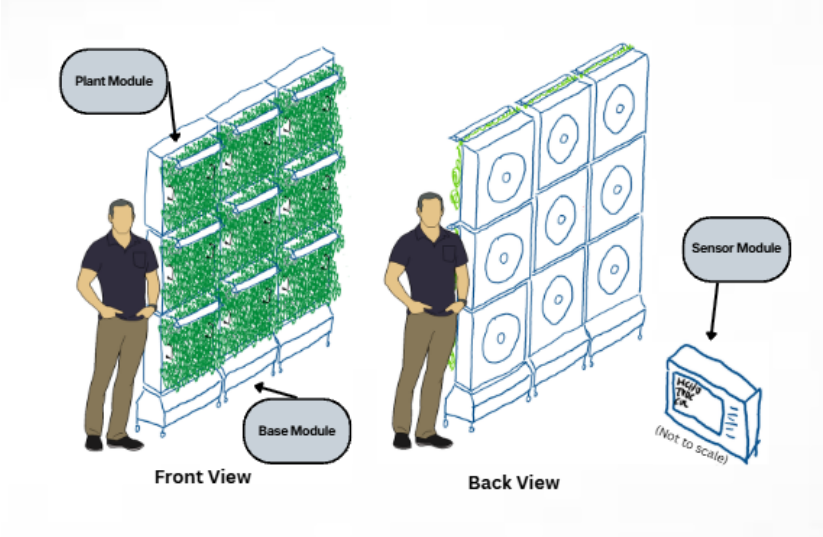

Figure 12: Overall Concept Design

Table 7: Subsystem Functions

Subsystem	Principal functions and components
Plant module	Accommodates the plants and growing medium, airflow chamber, fan, baffle, misting hardware and local lighting.
Base Module	A hybrid base houses the reservoir, pump and controller; supplementary reservoir bases increase water storage where required.
Sensor module	Measures selected indoor air-quality parameters, displays readings and communicates relevant information to the base controller.

The intended airflow path directs room air into the module chamber, distributes it across the planting face and passes it through the growing medium before returning it to the room. Sealing around the pots and holder is required to limit untreated bypass flow. The detailed inlet location and flow direction are shown in the engineering drawings.

Mechanical, water and electrical interfaces will be designed to support repeatable assembly and access for maintenance. Their detailed construction and connection methods are addressed in the Detailed Design section.

4.4. Concept Features Rationale

The proposed concept translates the design considerations in Table 4 into the features summarised below. These features establish the intended system functions while their effectiveness will be assessed during prototype testing.

Table 8: Features Rationale

Design Consideration	Proposed feature	How the feature addresses it
Modularity and Mobility	Repeatable plant modules and wheeled bases	• Modules can be combined horizontally or vertically, separated for transport and added or removed to suit spatial requirements. • Wheeled bases support repositioning when room layouts or preferred treatment locations change.
Visibility of air quality	Separate sensor module with an accessible display	• Displays pollutant readings and operating status so users can observe changing indoor conditions and understand the effectiveness of the plant modules.
Feedback capability	Threshold-based operating control	• Activates treatment fans at a predetermined setting when a selected control parameter exceeds its activation threshold. • Fans stop when the defined deactivation conditions are met.
Verified treatment capacity	Individually testable modules and configurable assemblies	• Supports measurements of airflow and purification performance, followed by tests with different module counts and room conditions. Results will inform sizing guidance within the tested range; experimental details are presented in Section 6.

5. Detailed engineering design and development

5.1 Plant-module construction

Each module has a 600 mm × 600 mm planting face and a 250 mm-deep enclosure, as shown in Figure 13. Thirteen 80 mm-diameter pots are arranged in alternating rows of three and two. A rear fan supplies air through an internal chamber and perforated baffle before it passes through the planted pots. Module-level lighting is supported in front of the planting face; its projection is additional to the enclosure depth.

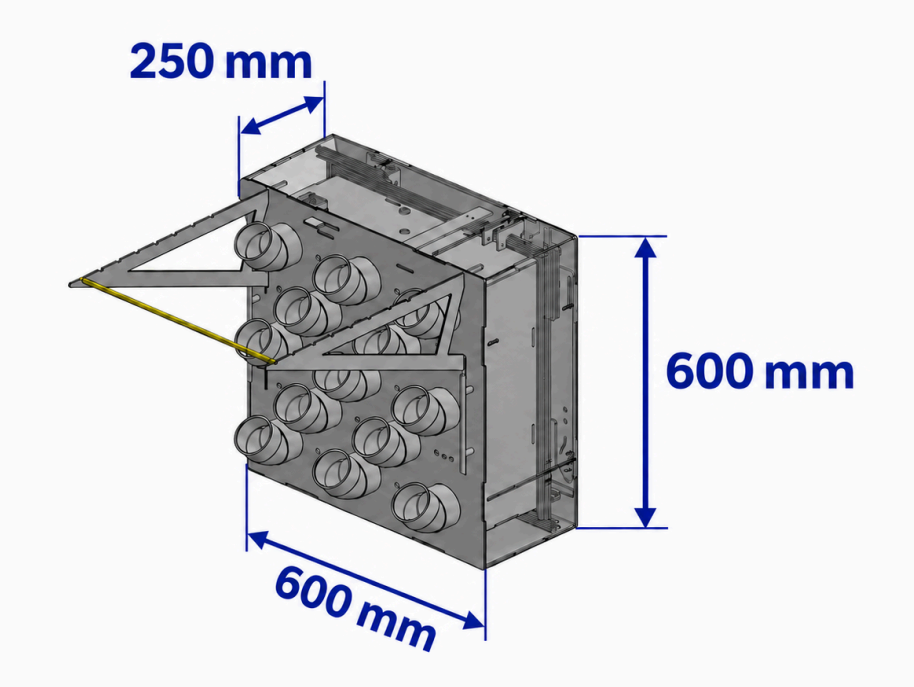

Figure 13: Plant Module Model

Construction combines laser-cut and 3D-printed components. Prototype assembly will verify pot fit, sealing, maintenance access and load transfer through the supporting structure under wet operating conditions. Development of the inter-module connection, including linkage and

rack-and-pinion motion transfer and alternative interlocking interfaces, is documented in Appendix 4. The estimated module mass (derived from SolidWorks) is 17.5 kg excluding plants. Component-level mass accounting and wet-module weighing will establish whether the complete module meets the provisional 25 kg handling target.

5.2 Irrigation design

The irrigation system supplies water from the base reservoir through a shared pump and inter-module pipe connectors and pipework to misting nozzles within each plant module as seen in Figure 14. The staggered pot arrangement provides access for mist distribution, with watering uniformity to be verified experimentally.

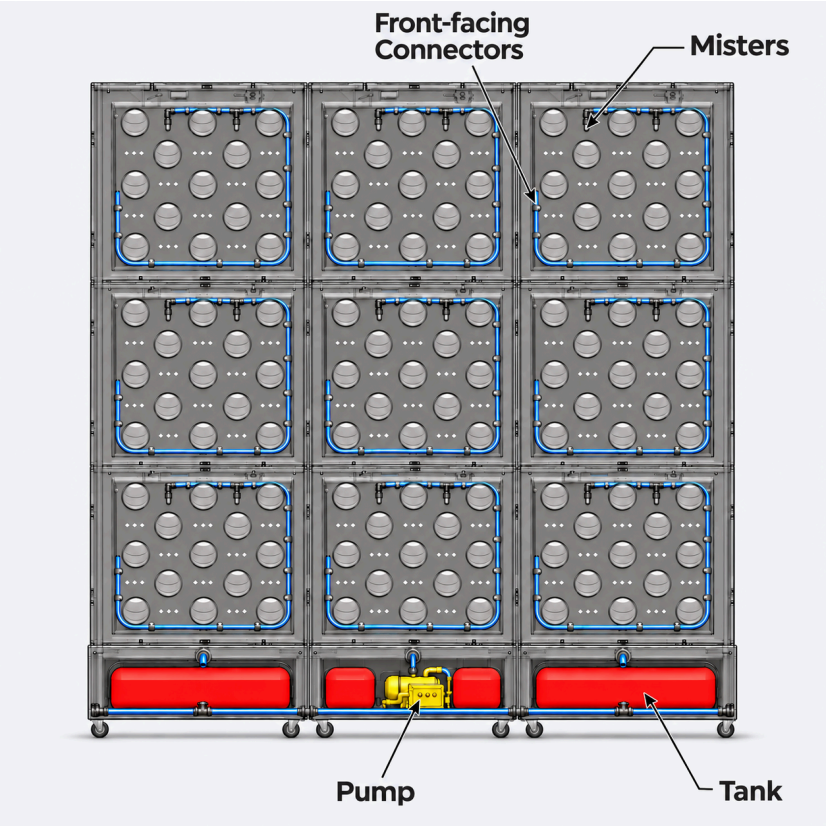

Figure 14: Irrigation Layout

Pump selection will account for delivery pressure, nozzle demand and the number and elevation of connected modules. Usable reservoir capacity will be assessed against measured consumption and losses to evaluate the 30-day refill target. Drainage, flow management, leakage detection and reservoir monitoring remain under development.

Calibration will establish nozzle flow, misting duration and cycle intervals. Airflow tests will use defined substrate moisture conditions, with irrigation timing, temperature and relative humidity recorded. Outlet velocities will be checked across the expected operating conditions against the provisional plant-airflow limit.

Net water consumption will be measured under the intended irrigation schedule to size the usable reservoir capacity against the 30-day refill target. As of yet, pump and tank sizing remains open to find the most suited for our use case.

5.3 Airflow and treatment-performance considerations

SPRE measures the proportion of a specified pollutant removed during one passage through the module. Single-pass clean air delivery rate (CADR) combines this efficiency with measured airflow, as defined in Appendix 2. Device-level measurements will account for leakage and bypass flow. Single-pass CADR characterises treatment through the module, while effective room treatment rate will be determined from the room experiments under defined baseline and operating conditions.

Figure 15 illustrates the potential trade-off between airflow and removal efficiency: lower airflow increases contact time, while higher airflow increases the volume processed. Experiment 1 will identify the tested operating setting that maximises CADR within the plant-airflow, moisture,

noise and power constraints. The illustrated relationship is conceptual and will be evaluated experimentally.

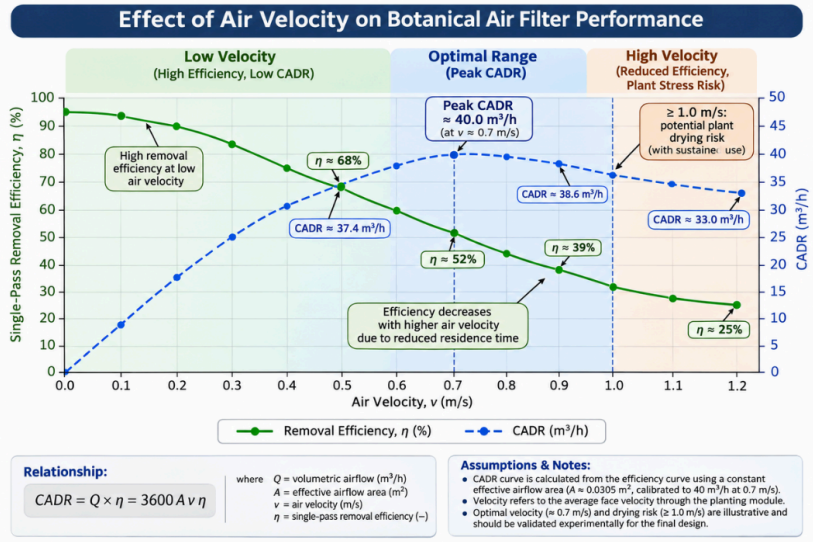

Figure 15: Illustrative graph demonstrating SPRE vs Air Velocity

A provisional assembly-level CADR benchmark of 350–400 m³/h was informed by particle air-cleaner sizing approaches. NEA recommends a minimum smoke CADR equivalent to five times the room volume per hour (NEA, 2024). For the largest intended room of 300 ft² (27.87 m²), an assumed ceiling height of 2.6 m gives a corresponding requirement of approximately 362 m³/h (27.87 × 2.6 × 5), within the adopted range. The AIRE calculator similarly gives approximately 350 m³/h using a rounded floor area of 27 m² and the same height and air-change rate (AIRE Purifiers, n.d.). This benchmark is a provisional assembly-level design reference, not demonstrated performance or a universal target for gaseous pollutants. Treatment capacity and the required module count will be established through pollutant-specific testing.

5.4 Baffle development and CFD findings

Three airflow configurations were compared in Ansys: no baffle and two perforated baffle designs. Without a baffle, airflow concentrated near the fan axis. Version 1 reduced this concentration but produced higher corner velocities. Version 2 showed a more uniform distribution across the 13 pot openings and was selected for prototype evaluation.

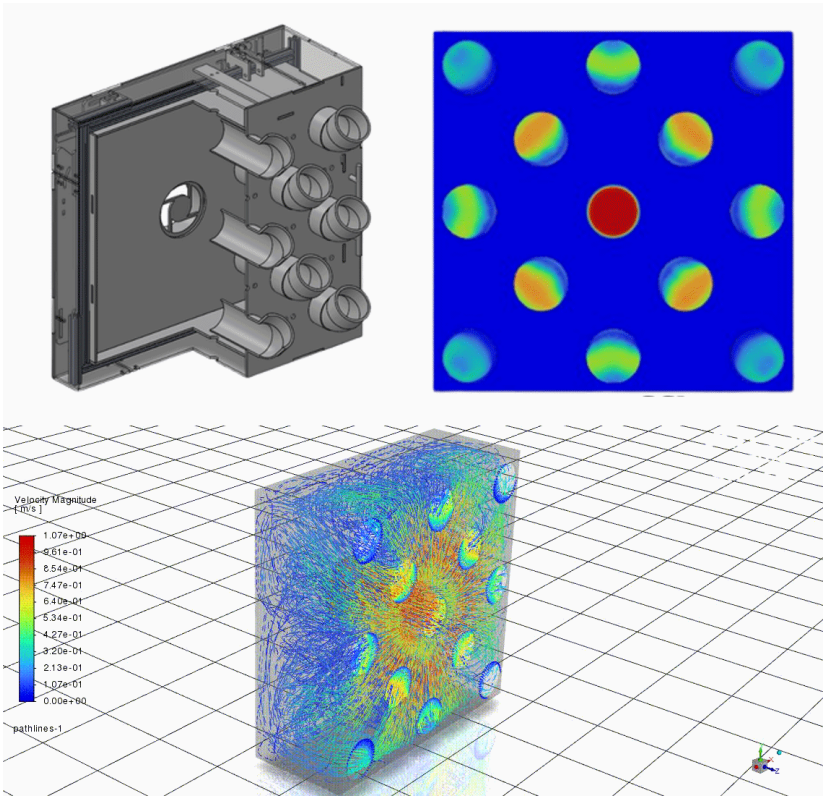

Figure 16: Ansys Simulation on No Baffle

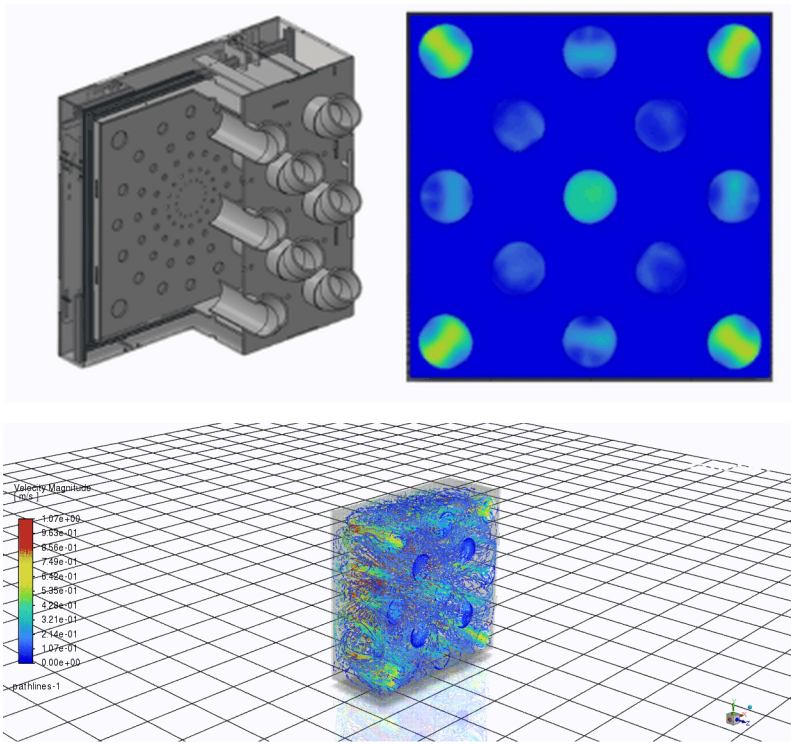

Figure 17: Ansys Simulation on Baffle Version 1

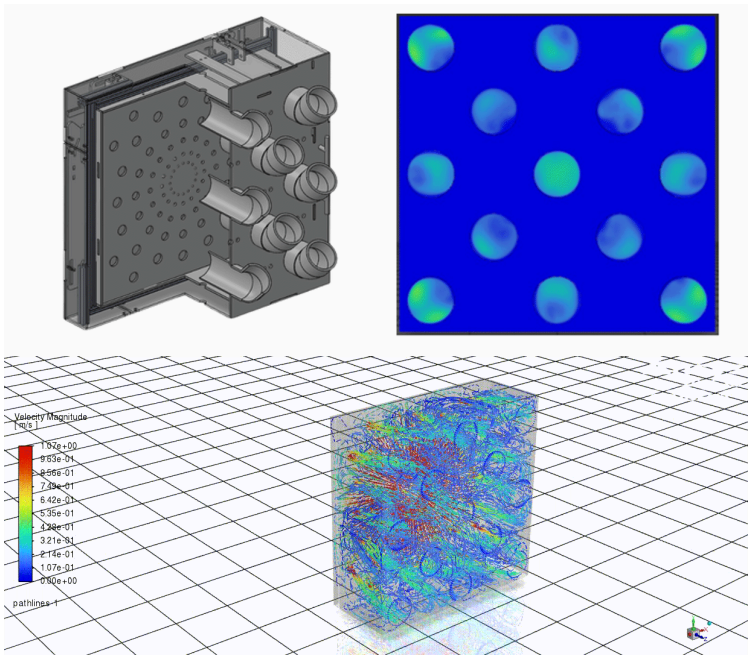

Figure 18: Ansys Simulation on Baffle Version 2

These findings remain qualitative. Quantitative assessment will compare area-averaged normal velocities across the 13 pot openings under consistent boundary conditions. Simulation assumptions are documented in Appendix 5, and prototype measurements will assess the predicted airflow distribution.

Simulated outlet velocities were approximately 0.5 m/s for a prescribed inlet velocity of 1 m/s. This comparison applies to the simulated geometry and boundary conditions; it does not establish a general 50% velocity reduction or pressure loss. Fan selection will therefore consider the assembled module's resistance and measured operating airflow.

5.5 Fan selection and control

The ebm-papst 4890 N was initially shortlisted as a candidate fan. Final selection will consider its pressure-flow characteristics against the assembled module's resistance and compatibility with the adjustable airflow required for Experiment 1. An alternative fan will be selected if these requirements cannot be met.

Calibration will vary the fan's operating setting using a compatible controller; pulse-width modulation (PWM) remains an option subject to fan compatibility. Actual airflow will be measured across all pot openings rather than inferred from the control setting. Pollutant removal, noise and power consumption will inform selection of the operating point.

Subsequent treatment experiments will maintain the selected setting continuously. During normal operation, configured pollutant thresholds will switch the fan on and off at this setting.

A plant-facing air-speed limit of 1 m/s remains a provisional constraint, with suitability to be verified for the selected plant and moisture conditions.

5.6 Lighting design and calibration

Module-level lighting is proposed to improve illumination uniformity across the assembly. Preliminary measurements from the reference installation indicate photosynthetic photon flux density (PPFD) values of 158 μmol/m²/s near the top and 22 μmol/m²/s near the bottom, supporting the use of distributed lighting. PPFD and daily light integral (DLI), including the calculation supporting the proposed lighting range, are explained in Appendix 6.

Figure 19: PPFD Distribution of NUS Flora Aer

The provisional lighting targets are a leaf-level PPFD of 40-80 μmol/m²/s, a daily photoperiod of 12–16 hours and a colour temperature of 4,000-5,000 K. The selected PPFD range is intended to support plant maintenance and moderate growth while limiting excessive foliage expansion and the associated need for pruning. These PPFD and photoperiod combinations correspond to a daily light integral of approximately 1.73-4.61 mol/m²/day. Suitability will be assessed through PPFD mapping and monitoring of plant condition.

Calibration will map PPFD across the 600 × 600 mm planting face. Lamp position and orientation will be adjusted to minimise uneven illumination and inter-module shading, with plant condition monitored during operation.

5.7 Base-module design

The base supports the assembly and houses shared services. Each assembly requires at least one hybrid base containing a water reservoir, pump and ESP32 controller. Additional water-tank bases increase storage capacity as the assembly expands horizontally.

Reservoir and pump sizing are addressed in Section 5.2. will depend on water demand, delivery pressure and module count. The design will provide maintenance access and separate water storage from electrical components. Structural testing will assess stability and wheel loading under filled and partially filled conditions to determine whether mobility can be retained.

5.8 Sensor module and integration

The proposed sensor module combines an ESP32 controller with the following I²C-capable sensors. Calibration and system integration remain pending.

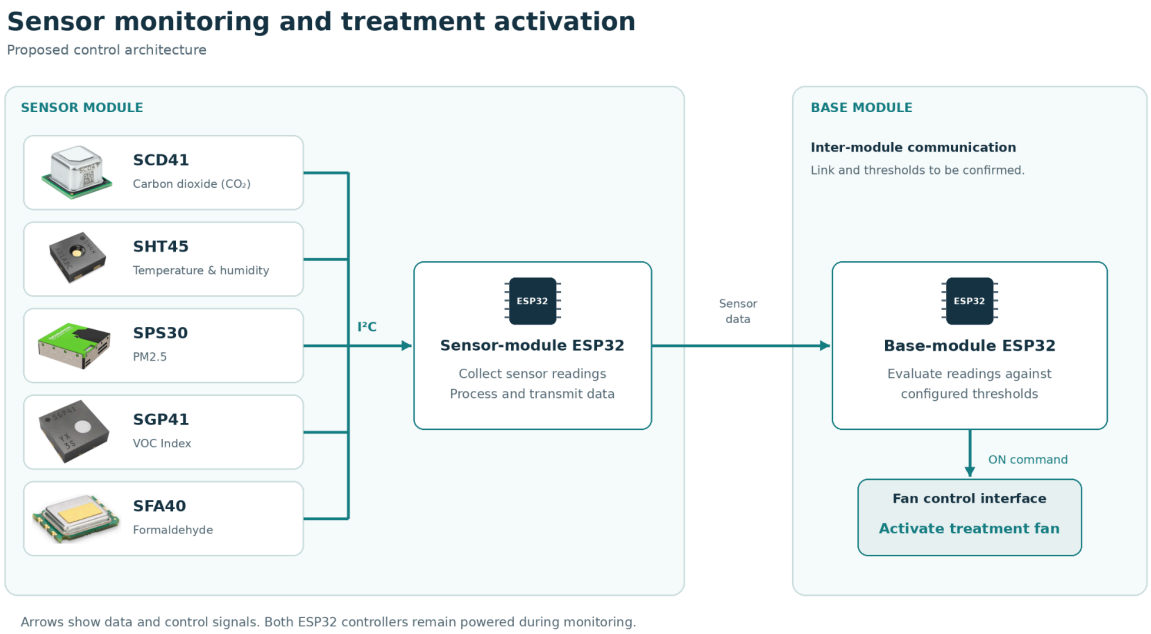

Figure 20: Sensor Selection and Proposed Control Architecture.

Figure 20 shows the selected sensors and their integration into the proposed control system. The sensor-module ESP32 collects readings and transmits them to the base-module ESP32, which activates the treatment fan according to configured thresholds. CO_2 provides indoor air-quality context (Sensirion, n.d.-a), while temperature and relative humidity help interpret plant conditions and sensor responses (Sensirion, n.d.-d). PM2.5 and formaldehyde measurements support air-quality monitoring (Sensirion, n.d.-b, n.d.-e), while the SGP41's signals are processed into a VOC Index for trend monitoring (Sensirion, n.d.-c).

Quantitative removal assessment must account for the measurement limitations of the SPS30 and SFA40 sensors (Sensirion, n.d.-b, n.d.-e). Potential mist interference will also be evaluated during testing. The VOC Index does not directly represent pollutant concentration (Sensirion, n.d.-c); quantitative VOC removal testing therefore requires a separate calibrated method. Similarly, CO_2 monitoring alone does not demonstrate removal by the system.

Integration will define communication between the sensor module and base, power supply, display, data logging and fault handling. Validation will include sensor co-location, timing alignment, reference comparisons and checks for interference from mist, humidity and outlet airflow.

6. Validation plan

6.1 Validation objectives and performance metrics

The validation programme will examine three design assumptions: that a suitable fan setting provides useful pollutant removal within operating constraints; that one module maintains effective treatment across different room configurations; and that adding modules increases treatment capacity predictably. A supplementary placement comparison will assess whether near-source treatment justifies retaining mobility.

Key measurements include pollutant concentrations, airflow, single-pass removal efficiency (SPRE), single-pass clean air delivery rate (CADR), effective room treatment rate and equivalent air changes per hour (eACH). Temperature, relative humidity, plant condition, noise and power consumption will establish the associated operating constraints.

The results will inform fan operation, supported assembly configurations and room-sizing guidance. SPRE and CADR definitions are provided in Appendix 2; experimental procedures and supporting calculations are detailed in Appendix 7, Appendix 8, Appendix 9, Appendix 10.

Prototype verification will additionally assess dimensions, wet mass, footprint, irrigation capacity, lighting and control response against the requirements in Table 5.

6.2 Experiment 1: Fan flow-rate optimisation

Experiment 1, seen in Figure 21, will identify the fan setting that provides the highest supported single-pass CADR within acceptable plant-airflow, moisture, noise and power limits. It will test

the proposed trade-off between airflow and removal efficiency and determine whether the selected fan and module provide a suitable operating point.

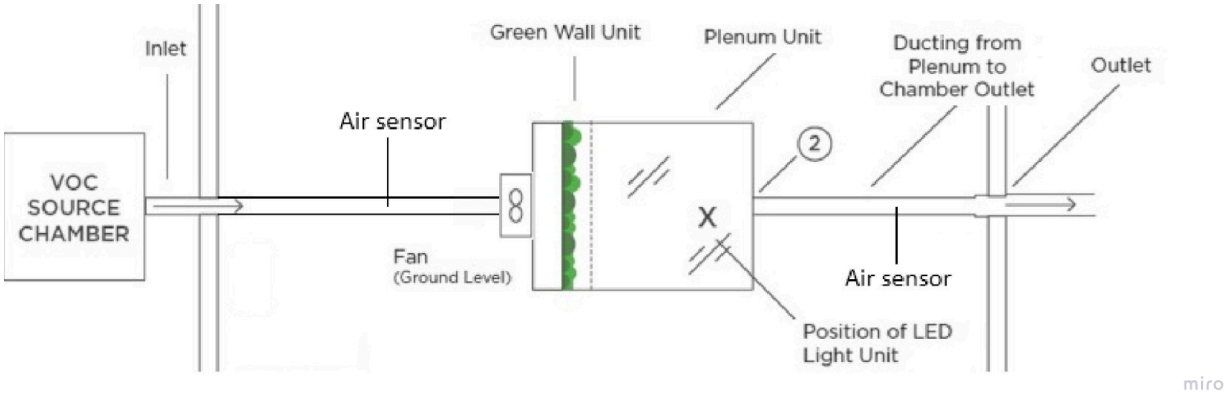

Figure 21: Experiment 1 Setup

VOC source selection will build on the TVOC and PM2.5 sources seen in Table 1, which include common adhesives, cleaning products, and burning mosquito coils. These materials will first undergo pilot assessment for measurement suitability and source stability. PM2.5 testing requires a separate particle source.

Representative inlet and outlet concentrations and measured airflow will be used to determine SPRE and CADR across fan settings. Noise, power consumption and plant operating conditions will also be recorded. Quantitative VOC assessment requires a calibrated concentration measurement method; VOC Index readings alone are insufficient.

The selected setting will be retained for Experiments 2 and 3. If no setting provides adequate treatment within the operating constraints, the fan, airflow path or treatment medium will be revised. Procedures, source qualification and calculations are detailed in Appendix 7.

6.3 Experiment 2: Room-volume analysis with a continuous source

Experiment 2, seen in Figure 22, will assess how the treatment provided by one module changes as room volume increases from V to 2V and 3V. It will determine whether a common single-module treatment rate can support room-sizing guidance and how the resulting equivalent air changes per hour decrease with increasing room size.

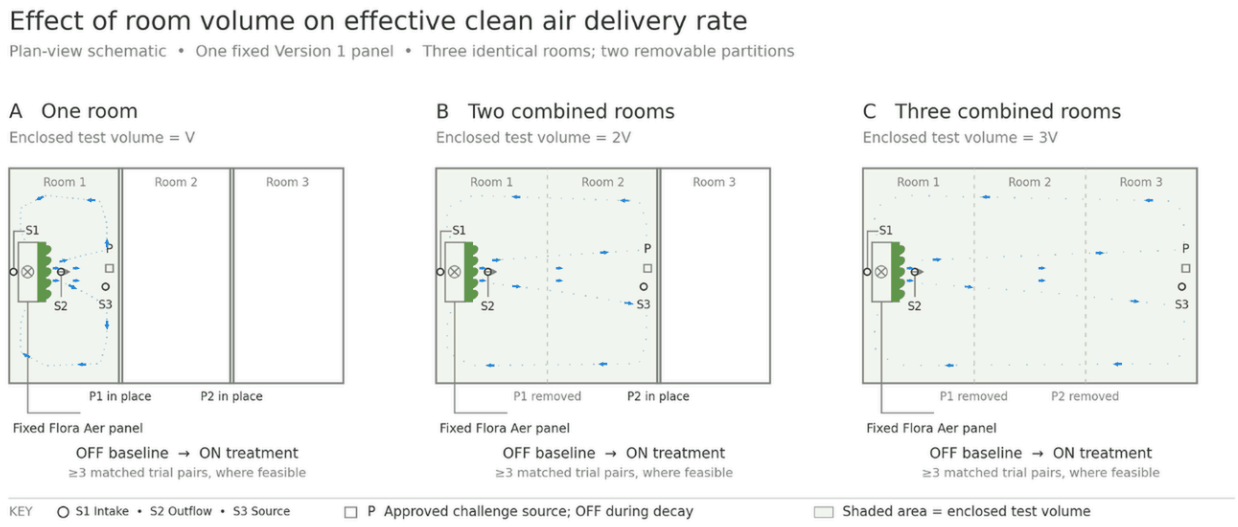

Figure 22: Experiment 2 Setup

Baseline and operating concentration histories will be measured near the source, at the module inlet and outlet, and within the occupied zone. Effective room treatment rate and equivalent Air Change per Hour (eACH) will be estimated using the continuous-source analysis in Appendix 8.

A consistent treatment rate would support a common module-capacity estimate within the tested range. If performance changes substantially, sizing guidance will account for room size and configuration. Because the source remains at the far end of each space, source-to-module

distance also increases; the results therefore represent room enlargement under this placement arrangement.

A supplementary fixed-volume placement comparison will assess whether near-source positioning provides sufficient benefit to justify mobility. Procedures, model assumptions and supporting derivations are provided in Appendix 8.

6.4 Experiment 3: Module-count and treatment-rate relationship

Experiment 3, seen in Figure 23, will assess the additional treatment capacity obtained by installing more modules in a fixed room. One, two and three modules will be installed sequentially, with every installed module operating at the setting established in Experiment 1.

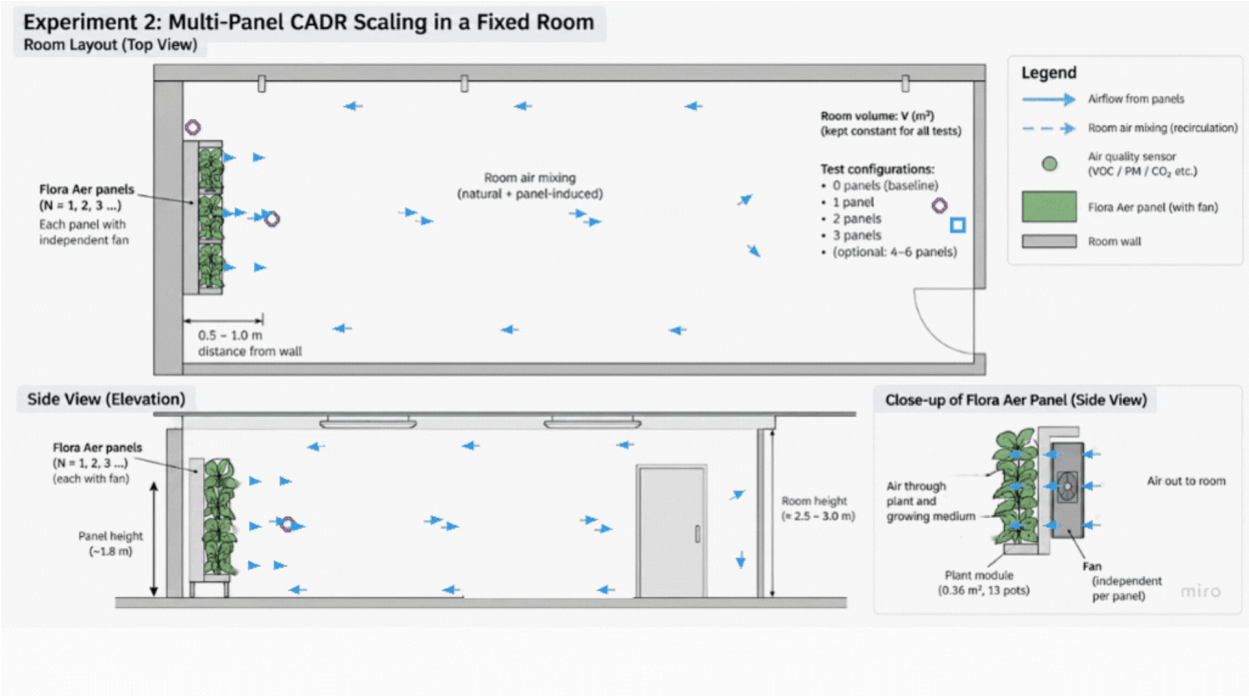

Figure 23: Experiment 3 Setup

Concentration histories, individual module airflow and installation arrangements will be recorded. Effective room treatment rates will establish the capacity gained from each additional module and whether performance scales approximately with module count.

The results will inform whether purchasing and installing another module provides a predictable increase in room treatment capacity. Reduced gains would prompt investigation of airflow imbalance, placement or interactions between modules. Recommendations will remain limited to tested configurations, with larger installations identified as projections. The module-count procedure and scaling-efficiency calculation are detailed in Appendix 9, using the continuous-source analysis derived in Appendix 8.

6.5 Development of the sizing guide

The sizing guide will plot pollutant-specific equivalent air changes per hour against room floor area, with separate curves for different installed module counts. This will show how treatment capacity per unit room volume changes with room size and how additional modules may increase it. An illustrative sizing chart seen in Figure 24 shows what we intend to plot using the findings from our 3 experiments to solve the uncertain room-scale performance design requirement seen in Table 4

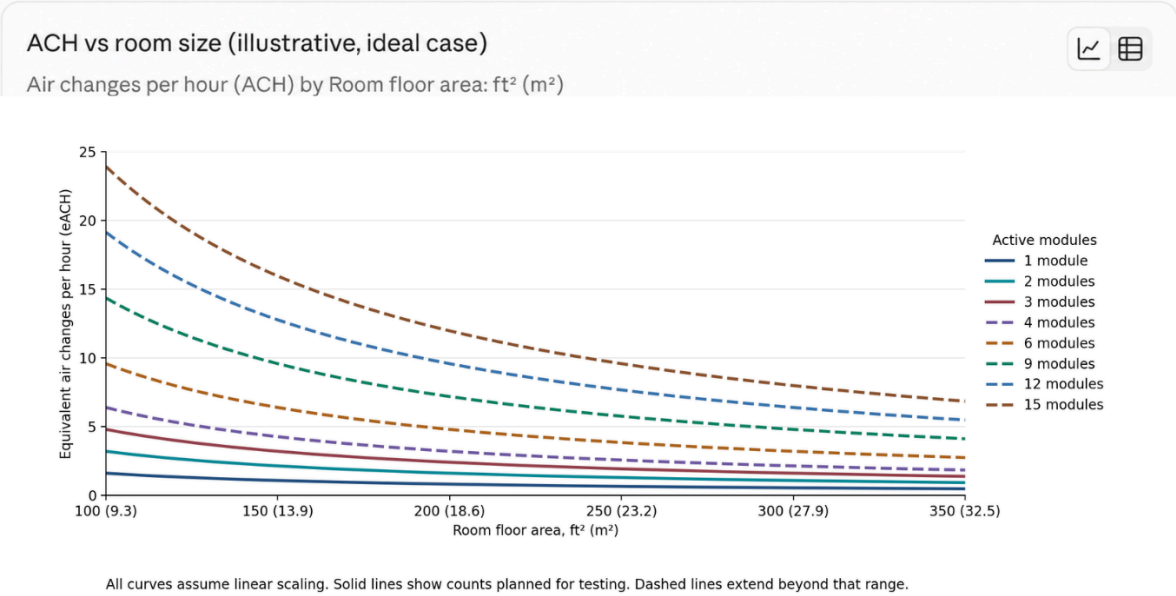

Figure 24: Illustrative Sizing Chart for Equivalent Air Changes versus Room Floor Area

Measured treatment rates for one, two and three modules will provide the initial basis for the guide. Floor area will be converted to volume using a stated ceiling height. Where treatment rates vary with room size or arrangement, the curves will reflect those findings rather than assume constant capacity.

Curves for four, six, nine, twelve and fifteen modules will remain projections unless subsequently tested. The present figure illustrates the intended presentation and assumes additive scaling; its values are not measured results or acceptance targets. Any recommended module count will also require a justified target eACH and compliance with structural, footprint and service constraints. Supporting calculations are provided in Appendix 10.

7. Progress and next steps

7.1 Prototyping progress

The project has progressed from concept exploration to detailed module design and initial fabrication. A flat modular configuration comprising plant, base and sensor subsystems has been selected. The plant-module design defines a 600 × 600 mm planting face, a 250 mm-deep enclosure and thirteen pot positions, with integrated airflow, irrigation and lighting provisions.

Development has included alternative interlocking mechanisms, qualitative CFD comparison of three airflow configurations and preliminary component selection. Baffle Version 2 was selected for prototype evaluation, while lighting targets and the sensor-control architecture have been defined. Sections 4–5 and Appendix 4 document these design outputs and their development.

Bearings and gears have been fabricated, while laser-cut panels remain under development. Full-module assembly and verification are outstanding. The next milestone is a functional prototype with documented fit, sealing, wet mass, airflow, irrigation and lighting performance. Treatment capacity and supported assembly configurations remain to be established experimentally.

7.2 Mapping progress to the objectives

The project addresses four objectives: flexible deployment, visibility of indoor air conditions, responsive operation and experimentally verified treatment capacity. These objectives guide the modular architecture, sensing and control functions, and validation programme. Table 9 maps current outputs to each objective and identifies the evidence needed to verify the principal design requirements.

Table 9: Progress against objectives and requirements

Objective or requirement	Current progress	Remaining verification
Deployment flexibility	Modular plant, base and sensor architecture defined	Assembly interfaces, stability and practical movement trials
Visibility of air conditions	Sensors and display concept selected	Verified readings, data logging and integrated display
Responsive operation	Threshold-based fan activation proposed	Communication, switching response and fault-handling tests
Treatment capacity	Experiments and analysis methods defined	Pollutant-specific single-pass, room-scaling and module-addition results
Dimensions and footprint	Enclosure dimensions defined	Complete installed and packed dimensions; assembly footprint
Module mass	Estimated at 17.5 kg excluding plants	Component mass accounting and complete wet-module weighing
Airflow and noise	Baffle selected qualitatively; fan candidate shortlisted	Installed airflow distribution and declared module/assembly noise measurements

Irrigation and refill interval	Shared supply and internal misting proposed	Water distribution, leakage, net consumption and reservoir sizing
Lighting	Module-level lighting and provisional targets defined	Leaf-position PPFD mapping and plant-condition monitoring

Current progress establishes the proposed design and its verification approach. Achievement of the objectives will depend on demonstrating integrated operation and measured performance within the supported configurations.

7.3 Outstanding technical questions

The principal measurement uncertainty is whether removal of the selected pollutants can be resolved reliably. Pilot tests will establish source stability, instrument suitability and measurement uncertainty. If quantitative VOC assessment is unsuitable, a calibrated reference method will be sought or the validated pollutant scope narrowed explicitly.

Installed airflow through wet growing medium and interference from misting also remain uncertain. Insufficient airflow will require revision of the fan, controller or flow path. Measurement interference will require changes to sampling arrangements or irrigation timing.

Wet mass, connection strength, base stability and water consumption will determine the permitted stack height, assembly configurations and reservoir capacity. Wheels will be retained only if repositioning provides a useful treatment benefit and the connected assembly can be moved practically.

Finally, room-scaling and module-addition tests will determine whether treatment capacity can be represented by a common sizing relationship. Where performance depends strongly on room

configuration or module arrangement, recommendations will be restricted to the conditions supported by the results.

7.4 Remaining work and project sequence

Table 10 summarises the remaining activities and their completion criteria, while Figure 25 presents their planned timing and overlap. A functional module and qualified measurement method are required before the main treatment experiments. Experiment 1 will establish the operating setting for Experiments 2 and 3. Base development, controls integration and additional-module fabrication can proceed in parallel, with three functional modules required before Experiment 3.

Table 10: Remaining Work

Stage	Remaining activity	Completion gate
Complete prototype development	Finish a functional plant module; establish seals, services and support; calibrate irrigation and lighting.	Assembled module with documented wet mass, operating flow and stable water and lighting performance.
Qualify the measurement method	Define challenge pollutants; verify source stability and sensor performance; document room and HVAC conditions.	Pilot data demonstrate that the expected removal can be resolved by the measurement system.
Run Experiment 1	Measure single-pass performance across fan settings.	Select an operating setting using measured CADR and operating constraints.

Run Experiments 2 and 3	Evaluate continuous-source room configurations and module-count scaling.	Repeatable data and a defensible estimate of effective treatment or clearly reported local concentration outcomes.
Evaluate mobility and integrate subsystems	Test placement benefit; complete base, service connections, sensor display and feedback control.	Decide whether wheels remain; demonstrate integrated operation.
Finalise the system and guide	Incorporate test findings, validate supported configurations and complete documentation.	Final prototype and a sizing guide with measured limits and labelled projections.

Experimental findings will inform final fan selection, supported assembly configurations and sizing recommendations. Changes to the fan or airflow path will require re-characterisation before subsequent performance comparisons.

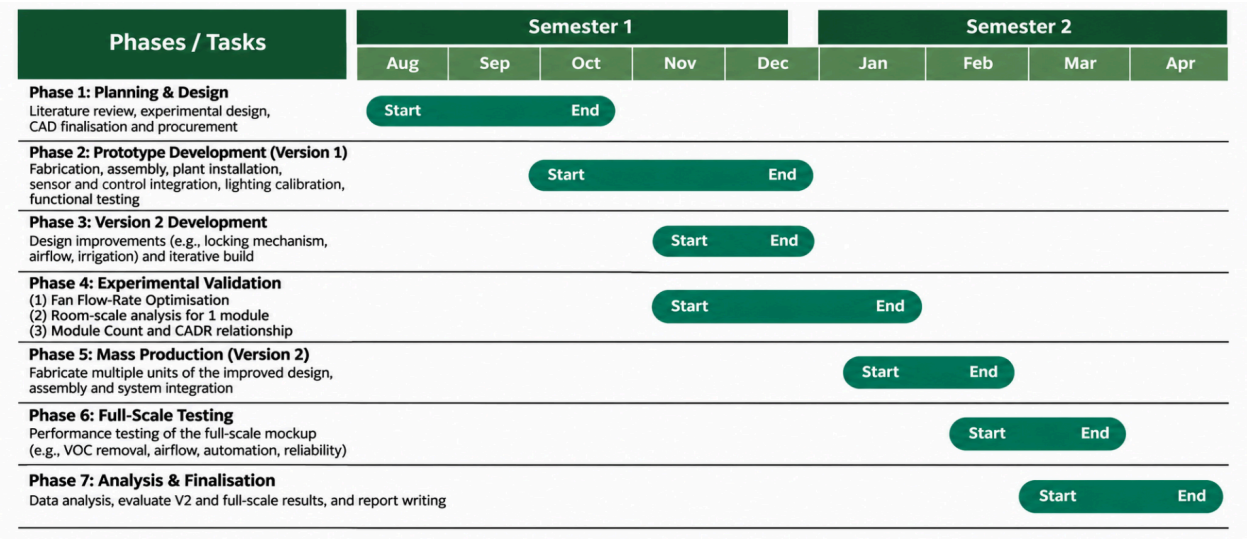

Figure 25: Gantt Chart

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Appendix

Appendix 1: Footprint Benchmark Using Naava One

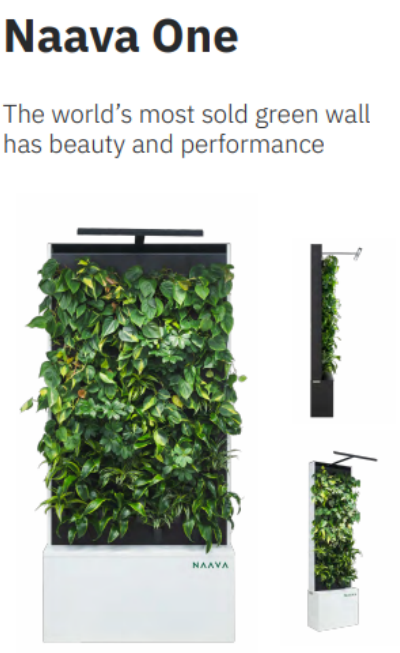


Figure 26 Naava One (Naava, n.d.)

Naava One is used as a commercial benchmark to assess the practicality of limiting the proposed assembly's base footprint to 5% of the smallest intended room area, 100 ft². The Naava brochure lists the following dimensions for its Naava One model: **1,003 mm width × 350 mm depth × 2,113 mm height** (Naava, n.d.). Only width and base depth are used for this calculation.

1. Room floor area

Using 1 ft² = 0.0929 m²:

Room floor area = 100 × 0.0929 = 9.29 m²

2. Naava One base footprint

   Base footprint = Width × Depth = 1.003 × 0.350 = 0.351 m²

3. Percentage of room floor area occupied

   Footprint percentage = (Base footprint ÷ Room floor area) × 100% = (0.351 ÷ 9.29) × 100% = 3.78%

The referenced Naava One model occupies approximately 3.78% of a 100 ft² room, demonstrating that a commercial botanical filtration unit can have a base footprint below 5% of this floor area. This provides a commercial precedent supporting the project-selected ≤5% assembly-footprint target, rather than establishing 5% as an industry standard.

This comparison covers the base footprint only. Plant and lighting projections, maintenance access and circulation clearances require separate consideration. It does not establish air-treatment suitability for a 100 ft² room.

Appendix 2: SPRE and CADR Definition

Single-Pass Removal Efficiency (SPRE) measures the percentage of a specified pollutant removed as air passes through the treatment system once.

$$SPRE\ (\%)\ = \frac{(Cin - Cout)}{Cin} \times 100$$

Where:

- Cin = pollutant concentration at the inlet.
- Cout = pollutant concentration at the outlet.

Both concentrations must use the same units and be measured under corresponding operating conditions. A higher SPRE indicates greater removal per pass, but does not account for the volume of air treated.

Clean Air Delivery Rate (CADR) expresses the equivalent volume of pollutant-free air delivered by the system per unit time. For a single-pass assessment, it combines measured airflow with removal efficiency:

$$CADR\ =\ Q \times \frac{SPRE}{100}$$

Where:

- Q = measured volumetric airflow through the treatment system (m^3/h).
- SPRE = single-pass removal efficiency expressed as a percentage.

SPRE and CADR are pollutant-specific and dependent on operating conditions. A higher SPRE does not necessarily produce a higher CADR if airflow is lower. Neither metric alone establishes

the percentage reduction achieved throughout a room, which also depends on room volume, ventilation, pollutant sources and air mixing.

Appendix 3: Flora Aer Reference-System Measurements

The team photographed and measured the existing Flora Aer prototype to establish a dimensional reference for the proposed design. Measurements covered the overall enclosure, planting arrangement, pot openings and lighting geometry. These observations informed initial space allocation, plant spacing and component positioning.

Figure 27 Flora Aer approximate dimensions

The observed pot spacing of approximately 18–19 cm provides a practical reference for the proposed 20 cm centre-to-centre spacing. Final dimensions will also account for fabrication, airflow, lighting and irrigation requirements.

Appendix 4 Development of the module interlocking mechanism

Appendix 4.1 Design objective

The interlocking mechanism was developed to connect neighbouring modules while permitting assembly and separation. Concept development addressed two related functions: transferring actuation around the module perimeter and engaging the connections between modules. Figure 28 and Figure 29 document the preliminary sketches, reference geometries and CAD exploration.

Appendix 4.2 Motion-transfer development

The first concept used perimeter linkages to transmit movement between adjacent sides of the module. The initial sketch and CAD model in Figure A8.1 show links arranged around the central opening, with pivoted connections at the corners. This explored coordinated movement of multiple connecting elements while retaining the central space for the module's other components. Joint clearances, available travel and potential interference remained to be assessed.

The second concept explored rack-and-pinion motion transfer. Figure 28 shows the proposed arrangement alongside a tooth-profile reference and a geometry calculation sheet. This approach investigated using meshing teeth to transfer movement between components. Further development requires confirmation of the rack guidance, gear mounting, travel and manufacturing tolerances.

The images record alternative motion-transfer concepts rather than verified mechanisms. Physical trials are required to compare operating force, alignment sensitivity and repeatability.

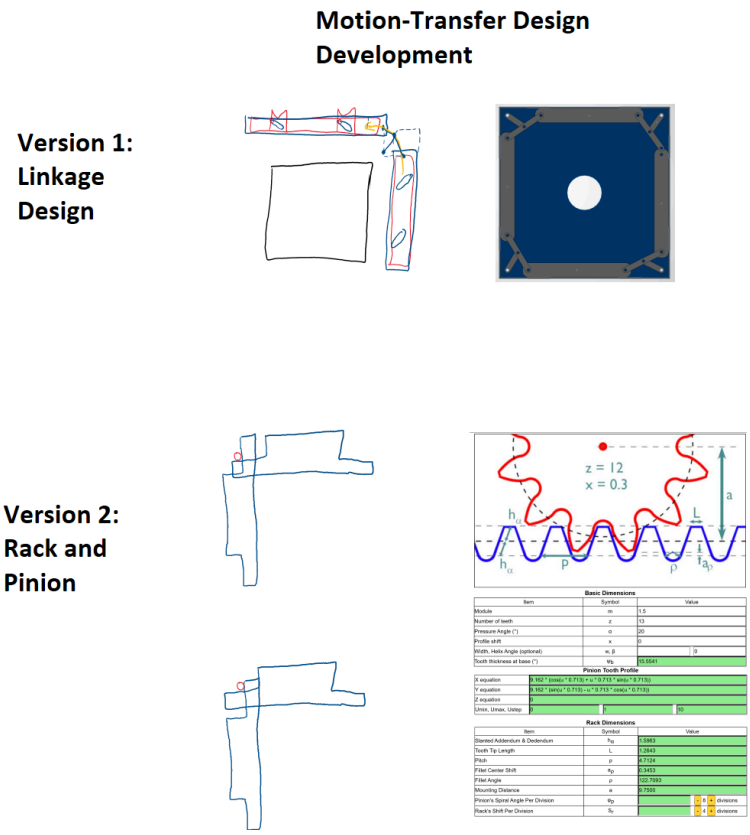


Figure 28: Motion-transfer development: Version 1 perimeter-linkage concept and CAD model; Version 2 rack-and-pinion sketches and supporting tooth-geometry exploration.

Appendix 4.3 Interlocking-interface development

The first interface concept explored an androgynous connection, in which matching interfaces could engage without separate male and female module types. Figure 29 shows connection features distributed around the module perimeter alongside an external reference for an androgynous coupling. The reference informed concept exploration; it does not establish the performance of the proposed module connection.

The second concept introduced alternating male and female features. The sketches show complementary features on neighbouring edges and sectional views exploring their engagement

through a guided sliding arrangement. This separates the locating and retaining features more explicitly, but requires compatible orientation and alignment between neighbouring modules.

The side and back views further explore how the connecting features could move within a housing. Their dimensions, clearances and integration with the motion-transfer mechanism remain to be resolved.

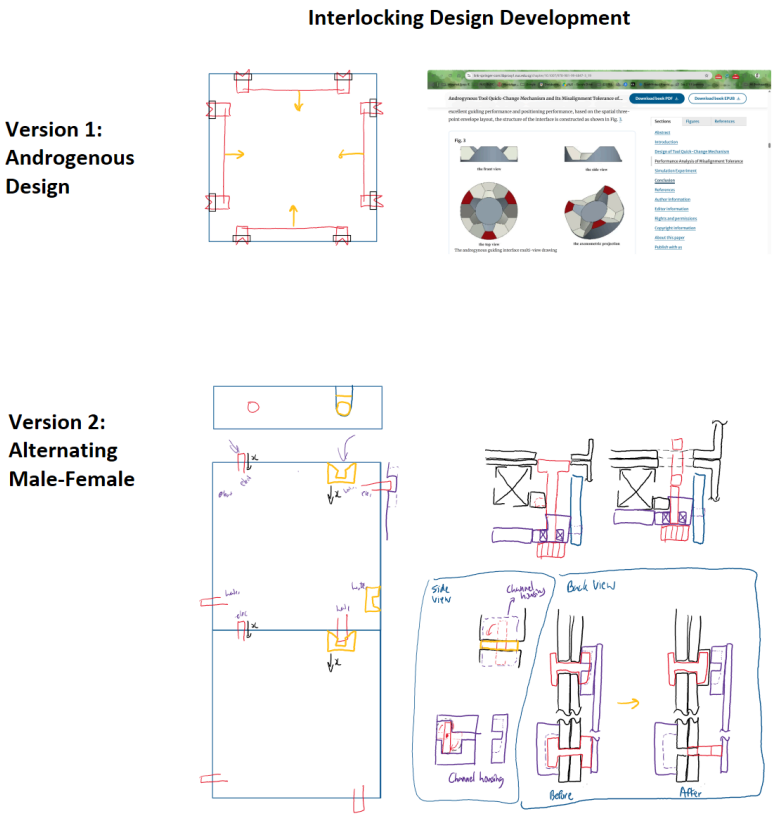


Figure 29 Interlocking-interface development: Version 1 androgynous concept and reference geometry; Version 2 alternating male–female arrangement and sectional engagement sketches.

Appendix 4.4 Further development and verification

The next stage will integrate the selected motion-transfer and interlocking concepts into a dimensioned assembly. Prototype checks will assess engagement and release, alignment tolerance, operating force, accidental disengagement and wear during repeated assembly.

Structural testing must establish how loads pass through the connections and supporting frame under wet operating conditions. Connection access and clearance must also be checked against irrigation, electrical services and maintenance requirements. These results will determine whether the mechanism is suitable for the proposed assembly configurations.

Appendix 5 Baffle Simulation Comparison

Three module airflow configurations were compared to investigate how a perforated baffle redistributes air across the planting outlets. The updated results support carrying Baffle V2 forward for prototype evaluation: its reported inner, middle and outer outlet-group velocities are more closely grouped, while the velocity contours show less central concentration than the no-baffle case. This remains a preliminary design selection pending numerical and physical verification.

Simulation Setup and Comparison Basis

The models were evaluated in Ansys Fluent 26.1 using a three-dimensional, steady, pressure-based solver with SST k–ω turbulence modelling. Air density was 1.225 kg/m³ and dynamic viscosity was 1.7894×10^{-5} kg/(m·s). Coupled pressure–velocity solution and second-order spatial schemes were used. No porous zone was enabled; resistance from the planted growing medium was therefore not represented.

The no-baffle and V1 reports specify a 1 m/s velocity inlet and zero-gauge-pressure outlets. V2 reports a 1 m/s inlet result, but its boundary-condition values are clipped in the PDF export and should be checked in the case file. Figure 31 uses matching displayed contour limits: approximately 0–1.07 m/s for velocity magnitude and −0.236 to 0.682 Pa for static pressure. These fixed display limits are not necessarily the minimum and maximum values throughout each model.

Outlet Distribution and Design Decision

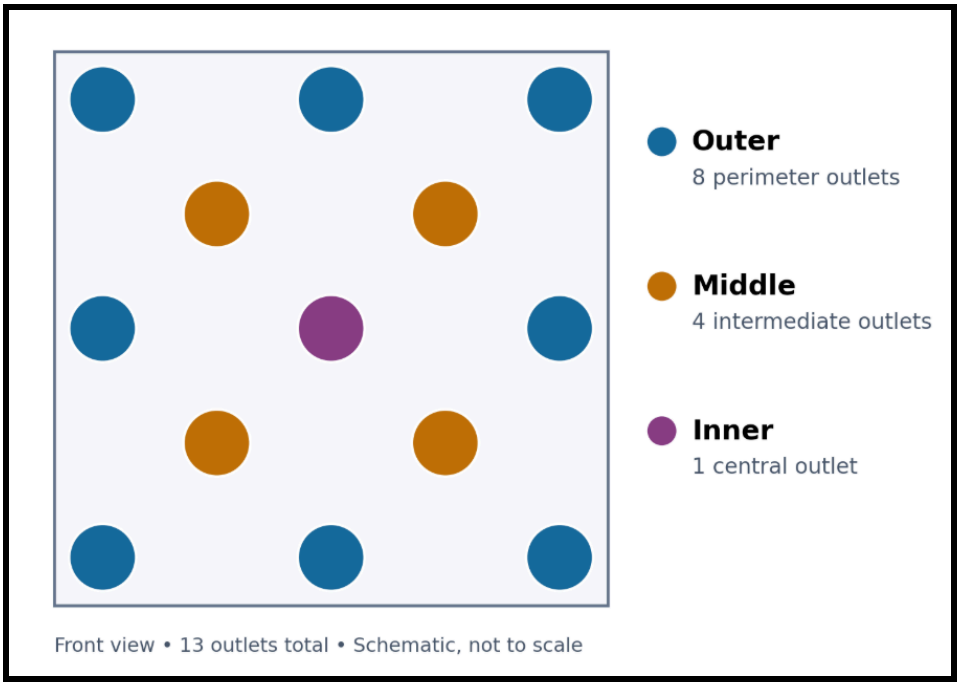


Figure 30 Outlet Grouping for Airflow Comparison

The 13 planting outlets were grouped by position: eight outer outlets along the perimeter, four middle outlets between the perimeter and centre, and one inner outlet at the centre. The same grouping was used across all three configurations to compare area-weighted mean velocity magnitudes and assess airflow distribution across the planting face.

Outlet group	No baffle	Baffle V1	Baffle V2
Inner	0.901	0.322	0.631
Middle	0.514	0.054	0.433
Outer	0.297	0.277	0.474
Group spread[1]	0.604	0.267	0.198

Table 11 Area-Weighted Outlet Velocity Magnitude (m/s)

[1]Highest minus lowest of the three group averages, calculated from unrounded values. This is a descriptive group comparison, not a 13-outlet uniformity index. Aggregate report totals are excluded because they also include the inlet.

No baffle: The central opening contains a pronounced high-speed region. The inner-group mean is approximately three times the outer-group mean, indicating central dominance in the reported result.

Baffle V1: The central concentration is reduced, but several intermediate openings remain dark blue while some peripheral openings carry higher-speed regions. The middle-group mean of 0.054 m/s indicates strongly uneven distribution between groups.

Baffle V2: More openings show appreciable velocity than in V1. The middle-group mean increases to 0.433 m/s and the three group means span 0.433–0.631 m/s. This supports carrying V2 forward to investigate more balanced delivery across the planting face; it does not establish uniform flow at every pot.

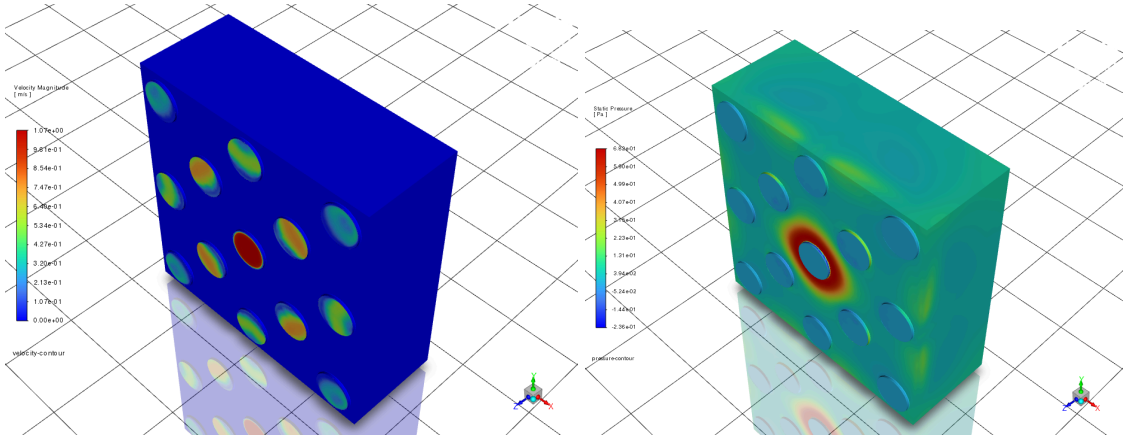
Status and Verification

All three exports record 50 iterations and “Not Converged”; continuity residuals are 0.1431, 0.2278 and 0.2179 for no baffle, V1 and V2 respectively. Before confirming the ranking, verify matched boundary conditions, convergence, mass balance and mesh sensitivity. Then measure flow distribution with the actual wet growing medium and seals. Velocity magnitude is not normal through-flow velocity; these results do not establish CADR, pressure loss or compliance at all leaf positions.

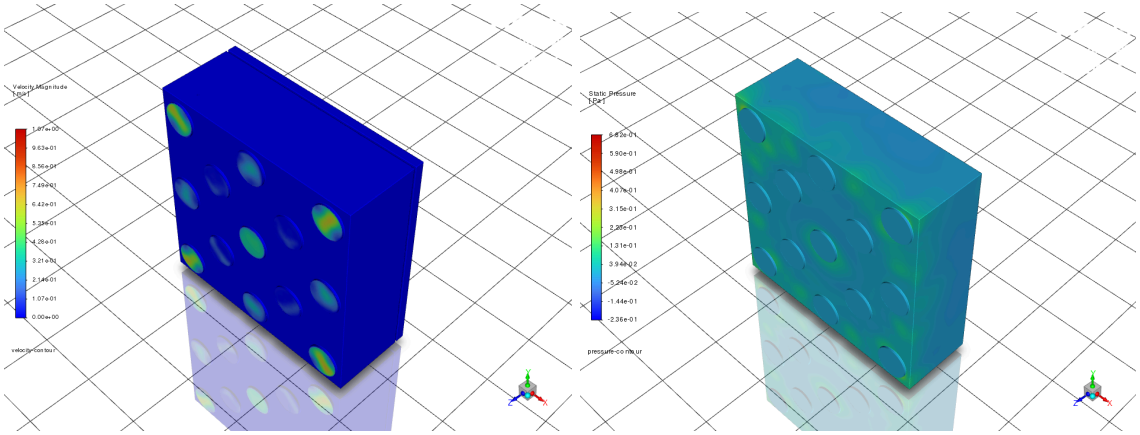
Velocity and Pressure Contours

Each row shows one configuration. Left: velocity magnitude (0 to 1.07 m/s). Right: static pressure (approximately −0.236 to 0.682 Pa). Original legends are retained.

No baffle



Baffle V1



Baffle V2

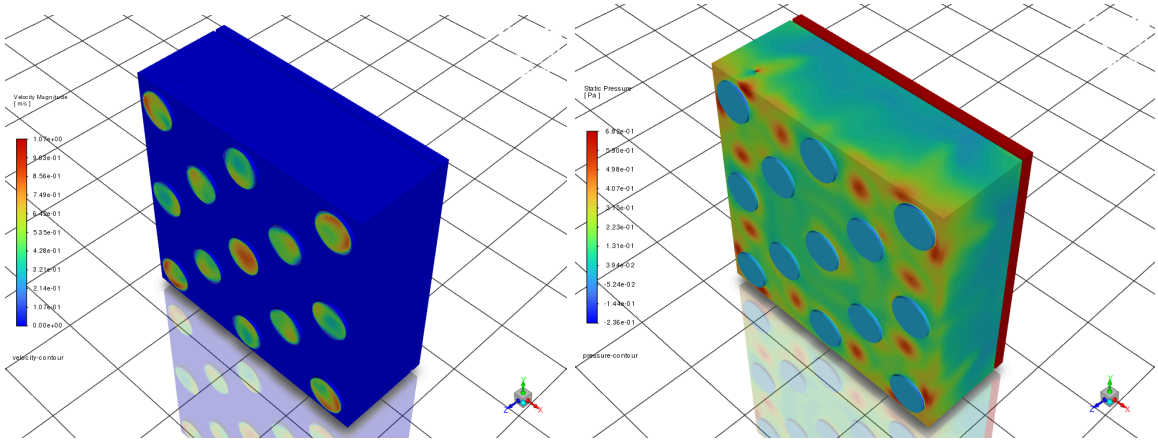


Figure 31: Matching-Scale Contours

The no-baffle case has a central velocity and pressure concentration. V1 suppresses the central velocity peak but leaves weak intermediate outlets. V2 shows more distributed outlet velocities and multiple local pressure regions. The V2 red regions indicate the upper displayed pressure band; they do not demonstrate a more uniform pressure field or quantify inlet-to-outlet pressure loss.

Internal Airflow Velocity lines

Velocity-line scenes combine the internal paths with the module geometry, using comparable transparent housing views. Line density depends on seeding and display settings and is not a measure of local airflow rate.

No baffle

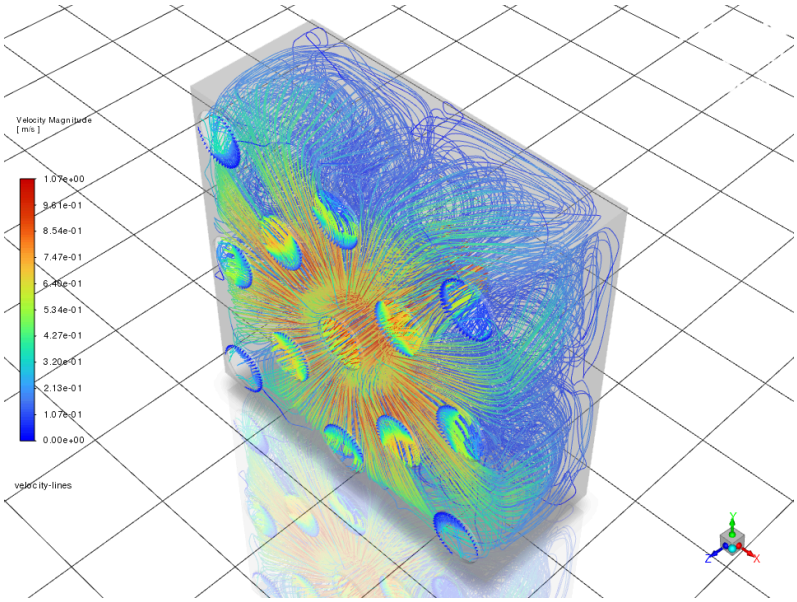


Figure 32: No Baffle Velocity Line

A prominent central high-speed region and curved internal paths accompany the centre-dominated outlet distribution.

Baffle V1

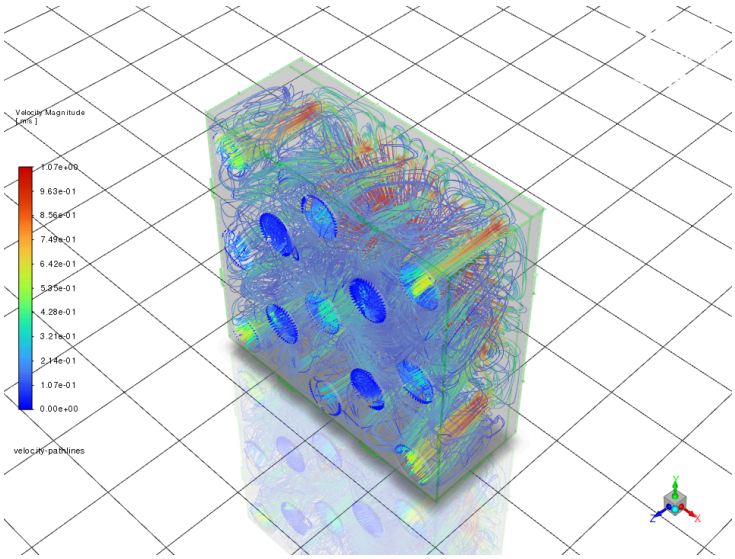


Figure 33: Baffle V1 Velocity Line

The baffle changes the internal routing, with higher-speed paths visible around peripheral regions. The outlet contour remains necessary to identify weakly supplied openings.

Baffle V2

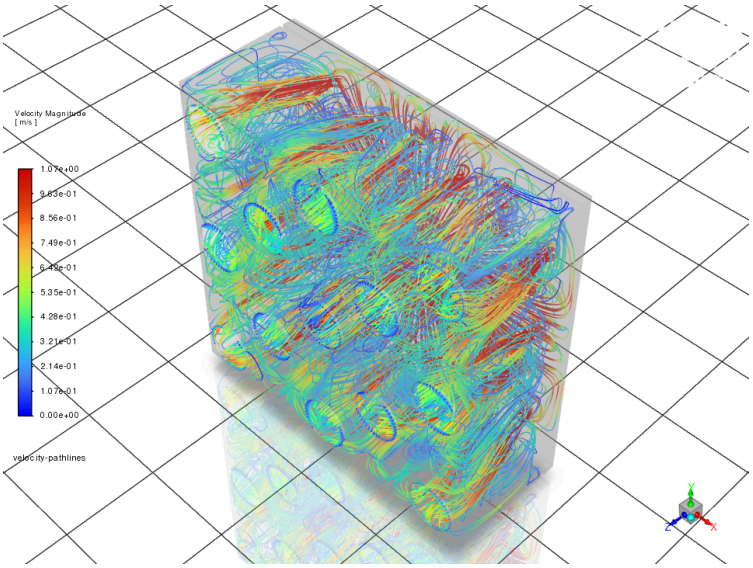


Figure 34: Baffle V2 Velocity Line

Paths extend across the internal chamber. Together with the outlet contours and group averages, the result supports further evaluation of V2 for more balanced outlet delivery.

Appendix 6 Lighting metrics and supporting calculations

Appendix 6.1 Photosynthetic photon flux density

Photosynthetic photon flux density (PPFD) measures the rate at which photons within the conventional photosynthetically active radiation range of 400–700 nm reach a surface. It is expressed in micromoles of photons per square metre per second (μmol/m²/s). PPFD therefore describes the light received at a measurement position, rather than the lamp's electrical power or colour temperature (Apogee Instruments, n.d.).

PPFD will be measured at representative leaf positions across the planting face using a suitable quantum sensor. Measurement position, sensor orientation and lamp distance will be recorded. Minimum, maximum and mean values will describe illumination distribution.

Appendix 6.2 Daily light integral

Daily light integral (DLI) is the total photosynthetically active photon exposure received per unit area over a day, expressed in mol/m²/day (Apogee Instruments, n.d.).

For constant PPFD during the illuminated period:

$$DLI = PPFD \times photoperiod \times 3600 / 1{,}000{,}000$$

Here, photoperiod is expressed in hours per day. The factors 3600 and 1,000,000 convert hours to seconds and micromoles to moles, respectively.

For the lower proposed combination:

$$DLI = 40 \times 12 \times 3600 / 1{,}000{,}000$$

$$DLI = 1.728 \approx 1.73 \text{ mol/m}^2\text{/day}$$

For the upper proposed combination:

$$DLI = 80 \times 16 \times 3600 / 1{,}000{,}000$$

$$DLI = 4.608 \approx 4.61 \text{ mol/m}^2\text{/day}$$

These values describe the proposed design range, not a verified biological optimum. Where daylight or lamp output varies, DLI must account for the changing PPFD throughout the day.

Appendix 7 Experiment 1 — Fan flow-rate optimisation

Appendix 7.1 Experimental procedure

Candidate challenge pollutants include formaldehyde, other selected VOCs and PM2.5. The final pollutants, emission methods and test facilities will be confirmed with the laboratory supervisor, considering the potential sources listed in Table 1. Each test will use a concentration measurement method suitable for the selected pollutant and test conditions. Formaldehyde removal will be assessed separately from other VOCs. VOC Index readings will support trend monitoring but will not be used to calculate SPRE or CADR.

1. Assemble one planted module with the intended fan, baffle, seals and growing medium.
2. Verify sensor agreement through co-location and reference comparison where available. Document instrument uncertainty and response time.
3. Establish a stable pollutant stream through the module, with representative upstream and downstream sampling.
4. Test selected PWM settings across the proposed 25–100% range.
5. After airflow and concentration signals stabilise, record inlet and outlet concentrations, actual airflow, temperature, relative humidity, electrical power and noise.
6. Conduct at least three independent runs per setting, subject to pilot-run variability and available facilities.

Substrate moisture, irrigation timing and plant condition will be documented. Sampling will represent the overall treatment flow rather than individual high-performing outlets. Leakage and bypass paths will be included in the device-level assessment.

Appendix 7.2 Airflow calculation

Airflow will be calculated from area-averaged velocities normal to the measured openings:

$$Q = 3600 \times \Sigma(u_i \times A_i)$$ **(Eqn Appendix 7.1)**

Where:

- Q = total measured airflow, m³/h.
- u_i = area-averaged normal velocity at opening i, m/s.
- A_i = area of opening i, m².
- Σ = summation across all measured openings.
- 3600 = conversion from seconds to hours.

Each opening will be sampled sufficiently to represent its velocity distribution. Leakage flows will be assessed separately where they are not captured by the measured openings.

Appendix 7.3 SPRE and single-pass CADR

The relationships defined in Appendix 2 are applied as follows:

$$\eta = (C_{i\square} - C_{ou\square}) / C_{i\square}$$ **(Eqn Appendix 7.2)**

$$SPRE\ (\%) = 100 \times \eta$$ **(Eqn Appendix 7.3)**

$$CADR_SP = Q \times \eta = Q \times SPRE / 100$$ **(Eqn Appendix 7.4)**

Where:

- $C_{i\square}$ = representative inlet pollutant concentration.
- $C_{ou\square}$ = representative outlet pollutant concentration.
- η = removal efficiency expressed as a fraction.
- CADR_SP = single-pass clean air delivery rate, m³/h.

Concentrations must refer to the same pollutant, use consistent units and represent corresponding operating conditions. Where outlet concentrations vary substantially, mixed-outlet sampling or flow-weighted concentrations will be required.

Appendix 7.4 Operating-point selection

SPRE and CADR_SP will be plotted against measured airflow. The selected setting will provide the highest supported CADR_SP within the provisional plant-airflow constraint and acceptable moisture, noise and power conditions.

Results will include repeat-run variability and measurement uncertainty. A unique optimum will not be claimed where differences between settings cannot be resolved. The selected setting will be retained for subsequent experiments unless design changes require re-characterisation.

Appendix 8: Experiment 2 — Room-configuration analysis with a continuous source

Appendix 8.1 Experimental procedure

Three adjacent spaces of equal volume will be combined progressively by removing partitions to create V, 2V and 3V. Actual dimensions and accessible air volumes will be recorded.

One module will remain at a fixed position. The pollutant source will be placed at the far end of each configuration, following the intended room-scaling arrangement. HVAC settings, furniture, doors and source preparation will be controlled and documented.

Each configuration will receive matched baseline and operating runs. During operating runs, the fan will remain continuously at the setting selected in Experiment 1, with automatic threshold switching disabled.

Concentrations will be recorded near the source, at the module inlet and outlet, and at a documented occupied-zone position. Recording will capture a resolvable transient towards equilibrium. Sampling duration and frequency will be established through pilot tests.

At least three independent runs per condition are proposed. Source stability and measurement suitability must be demonstrated over the fitted interval. Continued evaporation from a material will not, by itself, be treated as evidence of constant emissions.

Room enlargement changes both source distance and potentially air circulation. These effects will be considered when interpreting changes in effective treatment rate.

Appendix 8.2 Baseline definition

The primary baseline will retain the planted module in position with its treatment fan switched off. Plant condition, irrigation and lighting will remain matched to the operating run. The estimated treatment rate will therefore represent the additional effect of active operation relative to the inactive installation.

This comparison does not independently quantify passive removal or isolate the contribution of plants from the growing medium and other components. A separate absent-module comparison would be required to assess the complete installation against an empty-room baseline.

Appendix 8.3 Continuous-source mass balance

For a sufficiently well-mixed room:

Accumulation = Pollutant input − Pollutant removal

$$V \times dC/dt = G + Q_v \times C\square_u\square\square\square_\gamma - (Q_v + k\square \times V + D) \times C$$ **(Eqn Appendix 8.1)**

Where:

- C = representative room pollutant concentration, μg/m³.
- t = time, h.
- V = room volume, m³.
- G = indoor pollutant emission rate, μg/h.
- Q_v = effective ventilation flow, m³/h.
- $C\square_u\square\square\square_\gamma$ = supply-air pollutant concentration, μg/m³.
- $k\square$ = first-order background loss coefficient, h^{-1}.
- D = additional effective treatment rate relative to the selected baseline, m³/h.

Other concentration units may be used if all terms are converted consistently. The ventilation term assumes balanced supply and exhaust flows. Background losses may include deposition and passive uptake where these can reasonably be represented as first-order processes.

The model assumes approximately constant emissions, ventilation, supply concentration and removal coefficients over the fitted interval.

Appendix 8.4 Equilibrium concentration

At equilibrium:

$$dC/dt = 0$$

Substituting into Equation A4.1:

$$G + Q_v \times C\square_u\square\square\square_\gamma = (Q_v + k\square \times V + D) \times C\square\square$$

Therefore:

$$C\square\square = (G + Q_v \times C\square_u\square\square\square_\gamma) / (Q_v + k\square \times V + D)$$ **(Eqn Appendix 8.2)**

$C\square\square$ is the equilibrium concentration. With continuing pollutant input, this concentration is generally non-zero.

Appendix 8.5 Time-dependent concentration

Define the overall rate coefficient:

$$K = (Q_v + k\square \times V + D) / V$$ **(Eqn Appendix 8.3)**

Dividing Equation A4.1 by V and substituting Equation A4.2 gives:

$$dC/dt = -K \times (C - C□□) \textbf{ (Eqn Appendix 8.4)}$$

Separating variables and integrating:

$$\ln|(C(t) - C□□) / (C(0) - C□□)| = -K \times t$$

Hence:

$$C(t) = C□□ + [C(0) - C□□] \times \exp(-K \times t) \textbf{ (Eqn Appendix 8.5)}$$

Where:

- C(0) = concentration at the start of the analysed interval.
- C(t) = concentration at time t.
- K = rate coefficient, h^{-1}.
- exp(−K × t) = e raised to the power −K × t.

Equation A4.5 describes an increase or decrease towards equilibrium, depending on the initial concentration. If C(0) equals C□□, concentration remains constant. A source-off equation assuming decay towards zero is unsuitable while the pollutant source continues emitting.

Appendix 8.6 Effective treatment rate

For the baseline, additional active treatment is defined as D = 0:

$$K_off = Q_v / V + k□ \textbf{ (Eqn Appendix 8.6)}$$

With the module operating:

$$K_on = Q_v / V + k□ + D / V \textbf{ (Eqn Appendix 8.7)}$$

If ventilation and background losses remain unchanged:

$$K_{on} - K_{off} = D / V$$

Therefore:

$$D = V \times (K_{on} - K_{off}) \textbf{ (Eqn Appendix 8.8)}$$

Equivalent air changes per hour are:

$$eACH = D / V \textbf{ (Eqn Appendix 8.9)}$$

D is expressed in m^3/h and eACH in h^{-1}.

Appendix 8.7 Fitting and interpretation

Equation A4.5 will be fitted to suitable concentration histories to estimate K and C□□. Measurements must contain a resolvable transient; equilibrium measurements alone cannot independently identify K.

Model suitability will be assessed using spatial agreement, fitted residuals, source stability and repeatability. Strong concentration gradients, changing adsorption behaviour or insufficient sensor response may prevent a reliable room-wide estimate.

If the model is unsupported, location-specific concentration histories and reductions will be reported. If $K_{on} - K_{off}$ cannot be resolved within uncertainty, a reliable positive treatment rate will not be claimed.

Appendix 8.8 Supplementary placement and mobility assessment

Within one fixed room configuration, the module will be tested near the source and at a more distant position. Source output, HVAC operation, furniture and fan setting will remain consistent. Intake orientation will be documented.

Room-monitoring locations will remain fixed, while device-level inlet and outlet sampling will follow the module. Concentrations will be compared over a common observation period.

Time-integrated concentration is:

$$I_c = \int_0^T C(t)\, dt \quad \textbf{(Eqn Appendix 8.10)}$$

For discrete measurements, the trapezoidal approximation is:

$$I_c \approx \Sigma\{[(C_i + C_{i+1}) / 2] \times (t_{i+1} - t_i)\} \quad \textbf{(Eqn Appendix 8.11)}$$

Where T is the common observation duration. With concentration in μg/m³ and time in hours, I_c has units of μg·h/m³.

Lower integrated concentration indicates lower cumulative concentration at that monitoring location during the test; it does not independently establish room-wide CADR.

A movement trial will assess turning clearance, required operators, service disconnection and reconnection, leakage and stability. Treatment benefit and movement feasibility will jointly inform whether wheels are retained.

Appendix 9: Experiment 3 — Module-count and treatment-rate relationship

Appendix 9.1 Experimental procedure

The experiment will use a fixed room volume and progressively install one, two and three planted modules. All installed modules will operate continuously at the setting selected in Experiment 1. Module locations, spacing and orientation will be documented.

Source preparation, HVAC operation, room-monitoring positions and plant conditioning will remain consistent. Individual module airflow will be verified after each installation.

Each installed configuration will receive a matched fan-off baseline with the same planted modules present. This separates the additional effect of active operation from changes in passive surfaces introduced by adding modules. The baseline is a measurement control; module count is changed by physically adding modules.

Concentration histories will be collected for baseline and operating conditions, with at least three independent runs per condition proposed. Effective treatment rates will be calculated using Appendix 8 where its assumptions are supported.

An additional no-module reference may be used to compare occupied-zone concentrations before and after installation. Such comparisons will be reported separately from the active-treatment rates derived using configuration-specific baselines.

Appendix 9.2 Treatment-rate calculation

For each active module count:

$$D_{\square} = V \times (K_{on,n} - K_{off,n})$$ **(Eqn Appendix 9.1)**

Where:

- $D_{\square}$ = effective treatment rate with n active modules, m^3/h.
- V = fixed room volume, m^3.
- $K_{on,n}$ = fitted operating rate coefficient, h^{-1}.
- $K_{off,n}$ = corresponding baseline rate coefficient, h^{-1}.

A common baseline may be used only where the physical arrangement and background conditions remain sufficiently consistent.

Incremental capacity from additional modules

The additional active treatment capacity associated with installing the second and third modules is:

$$\Delta D_2 = D_2 - D_1$$

$$\Delta D_3 = D_3 - D_2$$

For fixed room volume V, the corresponding increases in equivalent air changes per hour are:

$$\Delta eACH_2 = (D_2 - D_1) / V$$

$$\Delta eACH_3 = (D_3 - D_2) / V$$

These quantities describe the measured gain from each installation step under the tested arrangement. Scaling efficiency remains:

$$\varepsilon_{\square} = D_{\square} / (n \times D_1)$$

Both incremental capacity and scaling efficiency will be interpreted with measurement uncertainty. Because D□ uses a matched inactive-installation baseline, it represents additional active treatment rather than the total passive and active effect of the installed system.

Appendix 9.3 Scaling efficiency

For a measured single-module treatment rate D_1, ideal additive performance predicts:

$$D_{□,ideal} = n \times D_1 \quad \textbf{(Eqn Appendix 9.2)}$$

Scaling efficiency is:

$$\varepsilon_{□} = D_{□} / (n \times D_1) \quad \textbf{(Eqn Appendix 9.3)}$$

For two modules:

$$\varepsilon_2 = D_2 / (2 \times D_1)$$

For three modules:

$$\varepsilon_3 = D_3 / (3 \times D_1)$$

Percentage scaling efficiency is:

$$\text{Scaling efficiency (\%)} = 100 \times \varepsilon_{□} \quad \textbf{(Eqn Appendix 9.4)}$$

Values near one, or 100%, indicate approximately additive performance. Departures will be examined alongside measurement uncertainty, source variability, individual airflow and module interactions.

Scaling efficiency will only be interpreted when D_1 is positive and sufficiently resolved. Testing up to three modules will not establish the performance of larger assemblies.

Appendix 10: Sizing-guide calculations and interpretation

Appendix 10.1 Room-volume calculation

For floor area A and ceiling height H:

$$V = A \times H \quad \textbf{(Eqn Appendix 10.1)}$$

Where:

- V = room volume, m^3.
- A = floor area, m^2.
- H = ceiling height, m.

Floor area expressed in square feet will be converted using:

$$A(m^2) = 0.092903 \times A(ft^2) \quad \textbf{(Eqn Appendix 10.2)}$$

Appendix 10.2 Equivalent air changes per hour

For measured effective treatment rate $D_{\square}$:

$$eACH_{\square} = D_{\square} / V \quad \textbf{(Eqn Appendix 10.3)}$$

Substituting V = A × H:

$$eACH_{\square} = D_{\square} / (A \times H) \quad \textbf{(Eqn Appendix 10.4)}$$

Where $D_{\square}$ is in m^3/h and $eACH_{\square}$ is in h^{-1}. These values represent equivalent treatment capacity, not outdoor-air ventilation.

If treatment rate remains constant:

eACH at volume V = D□ / V

eACH at volume 2V = D□ / (2 × V)

eACH at volume 3V = D□ / (3 × V)

Doubling or tripling room volume therefore reduces eACH to one-half or one-third respectively. Experiment 2 will assess whether the constant-treatment-rate assumption is supported.

Appendix 10.3 Required treatment capacity

For an adopted target eACH:

$$D_{required} = eACH_{target} \times V$$ **(Eqn Appendix 10.5)**

Or:

$$D_{required} = eACH_{target} \times A \times H$$ **(Eqn Appendix 10.6)**

The target must have a stated pollutant and application basis. Configuration recommendations will compare D_required with measured treatment rates and their uncertainty.

Appendix 10.4 Module-count projections

Where additive scaling is supported, a preliminary module-count estimate is:

$$n_{estimated} = \text{ceiling}(D_{required} / D_{1})$$ **(Eqn Appendix 10.7)**

Here, ceiling means rounding upwards to the nearest whole module.

This calculation assumes each additional module provides the same effective treatment rate as the first. Measured multi-module results will take precedence. Estimates beyond the tested range will be labelled as projections and checked against structural and service limits.

If Experiment 2 identifies substantial room-configuration effects, sizing recommendations will use configuration-specific treatment rates rather than one universal D_1.

Appendix 10.5 Limits of the sizing guide

The guide will state the tested pollutant, operating setting, ceiling height, room conditions, uncertainty and supported module arrangements.

Equal eACH does not guarantee equal equilibrium concentrations. As derived in Appendix 8:

$$C□□ = (G + Q_v \times C□_u□□□_\gamma) / (Q_v + k□ \times V + D)$$

Final concentrations also depend on pollutant emissions, incoming-air quality, ventilation and background removal. The guide will therefore describe measured treatment capacity without implying a guaranteed concentration or health outcome.